AIPOWER
Your AI Power Partner

CONNECTED SAFETY & RESCUE PLATFORM

SmartAlarm

From alerts to rescue action

One platform connecting people, equipment and responders — across industries and regions.



BEFORE • DURING • AFTER

SmartAlarm

See the threat. See who needs help.

An alert creates value when it leads to the right action, for the right person, in the right place.



People at risk	Where should I go? Who can help?
Control center	What is happening? Who is not yet safe?
Rescue teams	Where should we go? What skills are needed?
Family & caregivers	Has someone reached my loved one?

No SOS ≠ No need for rescue

SMARTALARM

CONNECTED SAFETY & RESCUE

GLOBAL SOLUTIONS FOR SAFER PEOPLE, BUILDINGS AND COMMUNITIES

1 DETECT



Multiple hazard detection

FIRE • STORM/FLOOD • ACCIDENT • FALL • SENSORS

2 UNDERSTAND



Incident understanding

AI-ASSISTED INSIGHTS • REAL-TIME SITUATION AWARENESS

3 GUIDE



Indoor & outdoor guidance

BEST ROUTE • VOICE DIRECTIONS • ACCESSIBLE OPTIONS

4 REACH EVERYONE



Multi-channel alerts

PA/VOICE • MOBILE ALERTS • DRONES • RESPONDER DISPATCH

5 FIRE PROTECTION



Automatic firefiighting & suppression

LOCAL CONTROL + VERIFIED FEEDBACK

6 RESCUE PROACTIVELY



Proactive assistance

SUPPORT AT RISK PEOPLE • TARGETED ASSISTANCE • FASTER RESPONSE

7 COORDINATE



Unified emergency coordination

BUILDINGS • SECURITY • FIRE • RESCUE • AUTHORITIES

8 CONFIRM SAFETY



Peace of mind for everyone

REAL-TIME STATUS • FAMILY UPDATES • ACCOUNTABILITY

THE COMPLETE SOLUTION • Detect → Understand → Guide → Reach → Rescue → Coordinate → Confirm

SmartAlarm

One coordinated response, seven checkpoints.

Each step answers a question so information stays connected across systems.

1	DETECT	What signal was received?	Camera • IoT • wearable • SOS
2	UNDERSTAND	Who? Where? How severe?	Evidence • location • time
3	GUIDE	Where to go? What to do?	Maps • voice • accessible options
4	REACH	Who remains unreached?	App • PA • SMS • vehicles
5	PROACTIVE RESCUE	Who needs help?	Presence • assistance needs
6	COORDINATE	Who is responsible?	Assign • accept • hand over
7	CONFIRM	Who is actually safe?	Self-report • team verification • family

One safety core. Many operating environments.

Organizations and sites share the platform; each industry has its own data, workflows and integrations.

01 Buildings & campuses

Fire, evacuation and people accountability.

02 Industry & logistics

Hazardous areas and lone workers.

03 Care & communities

Falls, assistance needs and caregivers.

04 Transport & fleets

Collisions, road location and coordination.

05 Schools & events

Groups, assembly and guardians.

06 Cities & disasters

Affected areas, shelters and multiple agencies.

SmartAlarm

Prepare before. Coordinate during. Confirm after.

A continuous lifecycle, from site surveys to exercises and improvement.

01 BEFORE

Maps & equipment
Care profiles
Duty teams & exercises

02 DURING

Verify & alert
Guide & rescue
Track every task

03 AFTER

Confirm each person
Hand over & close
Learn from the response

Every status has an accountable owner, a timestamp and a source of confirmation.

GUIDED EVACUATION

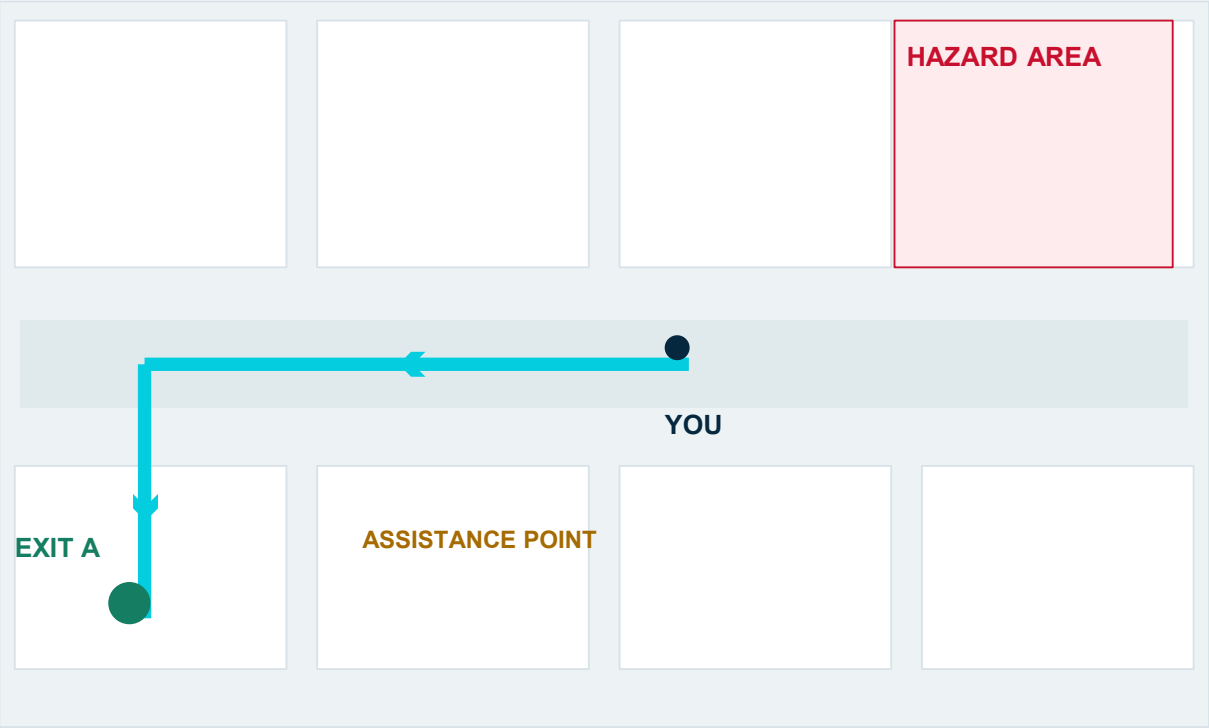


FIRE SCENARIO • Avoid hazardous areas • Step-by-step voice • Assistance for people who cannot self-evacuate

SmartAlarm

A fire on floor 5: from signal to personal contact.

Illustrative sequence — every action is linked to the same incident.



Conceptual diagram • not a site-approved evacuation plan

01 / VERIFY

Cameras and sensors mapped to floor and zone.

02 / GUIDE

Exclude hazardous routes; issue app and PA guidance.

03 / FIND PEOPLE

SOS requests plus proactive welfare checks.

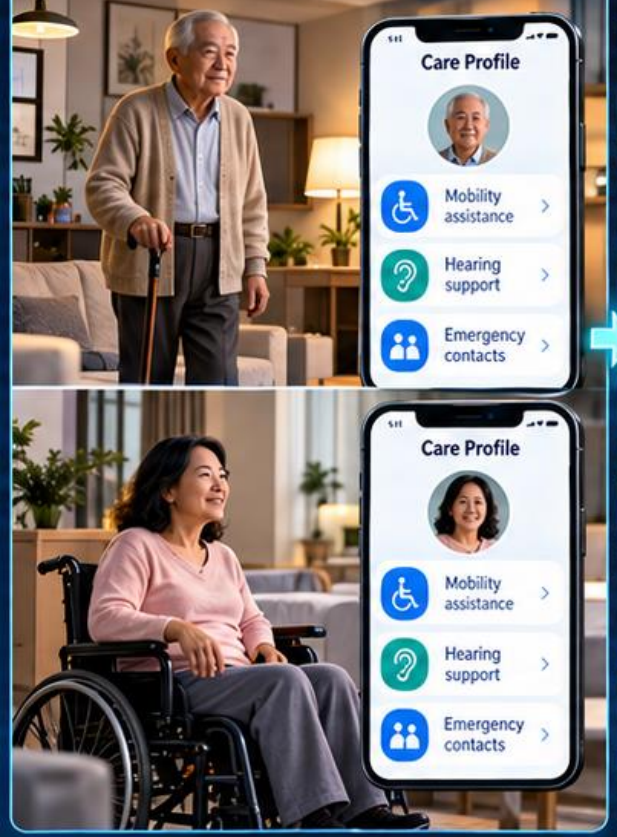
04 / RESCUE

Teams accept tasks, update status and hand over.

RESCUE WITHOUT SOS

1 PREPARE

Know people, needs and contacts before an emergency.



2 IDENTIFY

Detect hazards and locate people in real time.



3 COORDINATE

People review evidence and assign trained responders.



4 VERIFY

Responders assist to safety and notify families.



CARE PROFILE → EXPOSURE → PRESENCE → HUMAN REVIEW

PROACTIVE RESCUE • Assistance needs + Exposure + Presence + Operator assessment

SmartAlarm

Understand assistance needs, beyond a person's name.

Profiles retain their source and sharing permissions and are used for the intended rescue purpose.

CARE PROFILE

Minh Anh

Mobility assistance required
Audio guidance preferred
Emergency contact: verified
Last known location: floor 5
Data freshness: check required

Purposeful data	Age, mobility, hearing/vision and rescue-relevant conditions.
Reasoned priority	Exposure + presence + ability to self-evacuate + condition.
Act without a tap	Call to check, involve a caregiver or dispatch a team.
Limited disclosure	Teams receive necessary details; family access follows permissions.

SmartAlarm

Reliable location is a foundation for rescue.

Match technology to the environment; always show time, floor and confidence.

OUTDOORS

GNSS / device location

Address, road and direction

Hazard areas and assembly points

INDOORS

QR / BLE / UWB / Wi-Fi

Building → floor → zone

On-site survey and calibration

WHEN UNCERTAIN

Select a landmark

Confirm floor / location manually

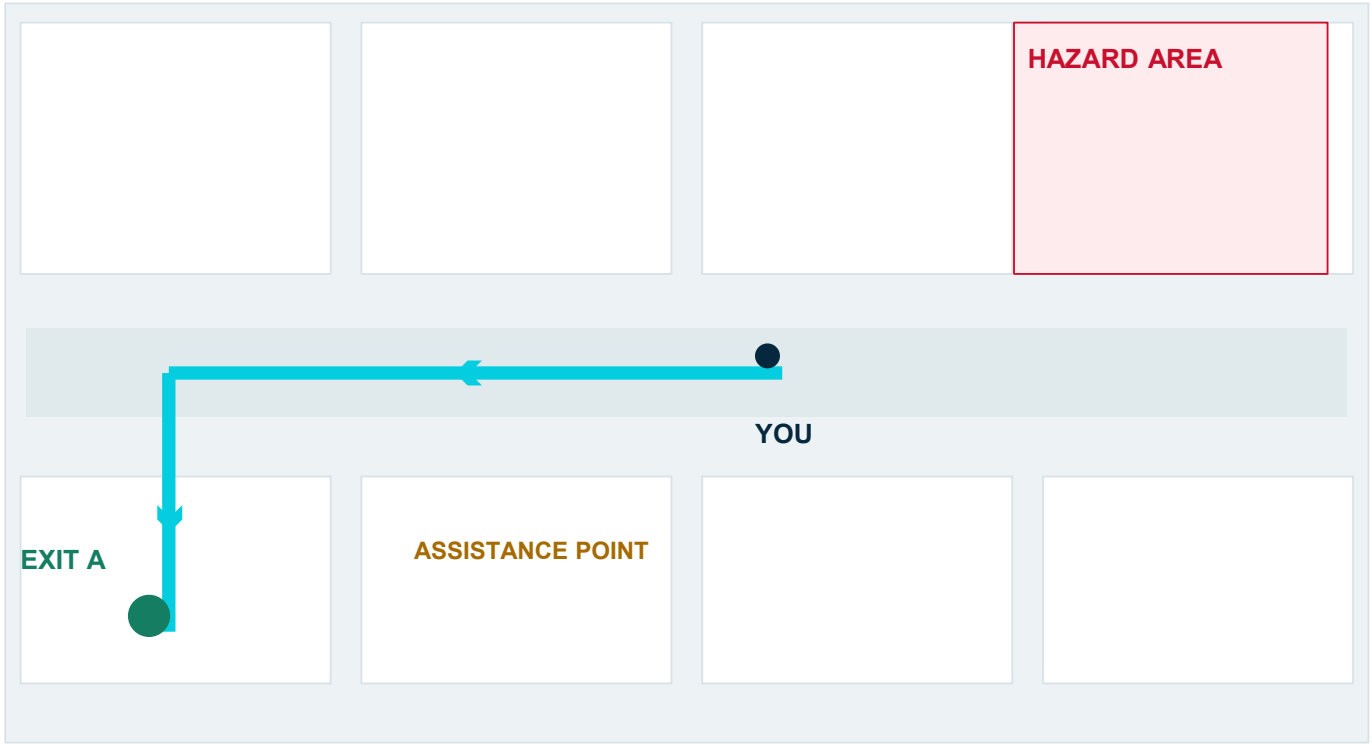
Assign a verification task

LOCATION + SOURCE + ERROR + FRESHNESS

SmartAlarm

The best route must be feasible and suitable.

Exclude hazards first, then optimize time, congestion and accessibility needs.



01 EXCLUDE UNSAFE ROUTES

Fire, smoke, locked doors, blocked paths and expired information.

02 SELECT A SUITABLE ROUTE

Mobility, passage capacity, confidence and estimated travel time.

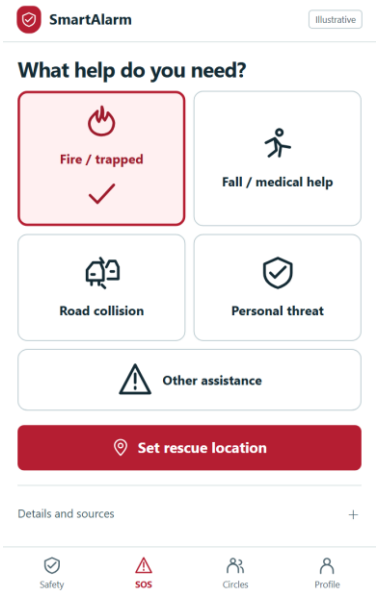
03 UPDATE AS CONDITIONS CHANGE

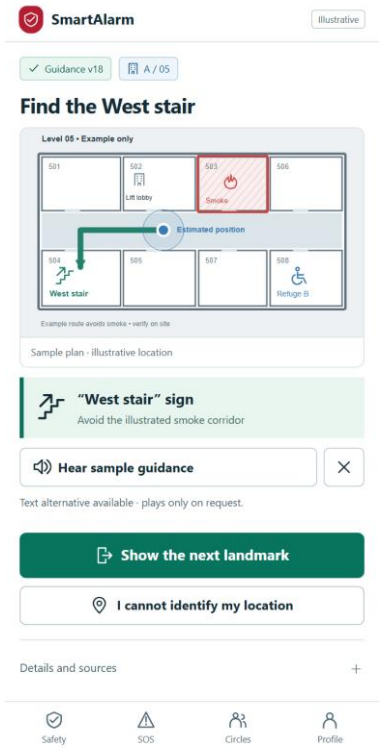
Withdraw the old route and voice; request assistance if no route remains.

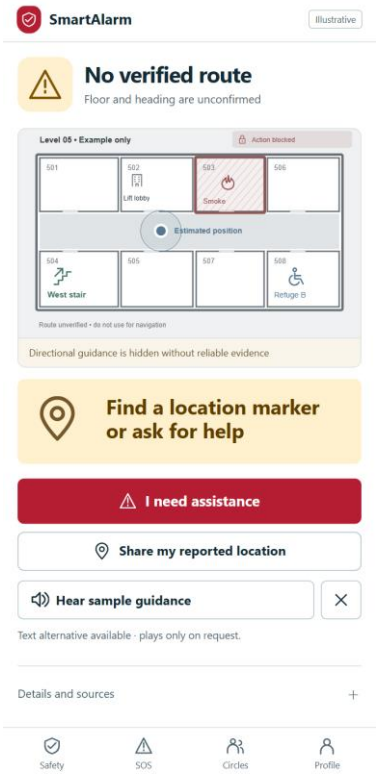
SmartAlarm

See the next action. Hear it when needed.

Visual guidance • recognizable landmarks • assistance within reach







SOS BY PICTURE
Choose the situation without a long form.

LANDMARK GUIDANCE
Map and next action stay together.

NO VERIFIED ROUTE
Withdraw arrows; offer assistance.

ONE MESSAGE. EVERY CHANNEL.



RESCUE AUDIO • Building PA + Smartphone voice + Drone speaker + Vehicle PA — one approved message

SmartAlarm

One message. The right area. The right version.

Track each delivery channel and escalate when the primary channel fails.

01 / DRAFT & APPROVE

Area, severity and language
Reviewed emergency templates
Approval by action type

02 / DISTRIBUTE

App / push / SMS / contact
Building / vehicle PA
Drone by supported capability

03 / TRACK

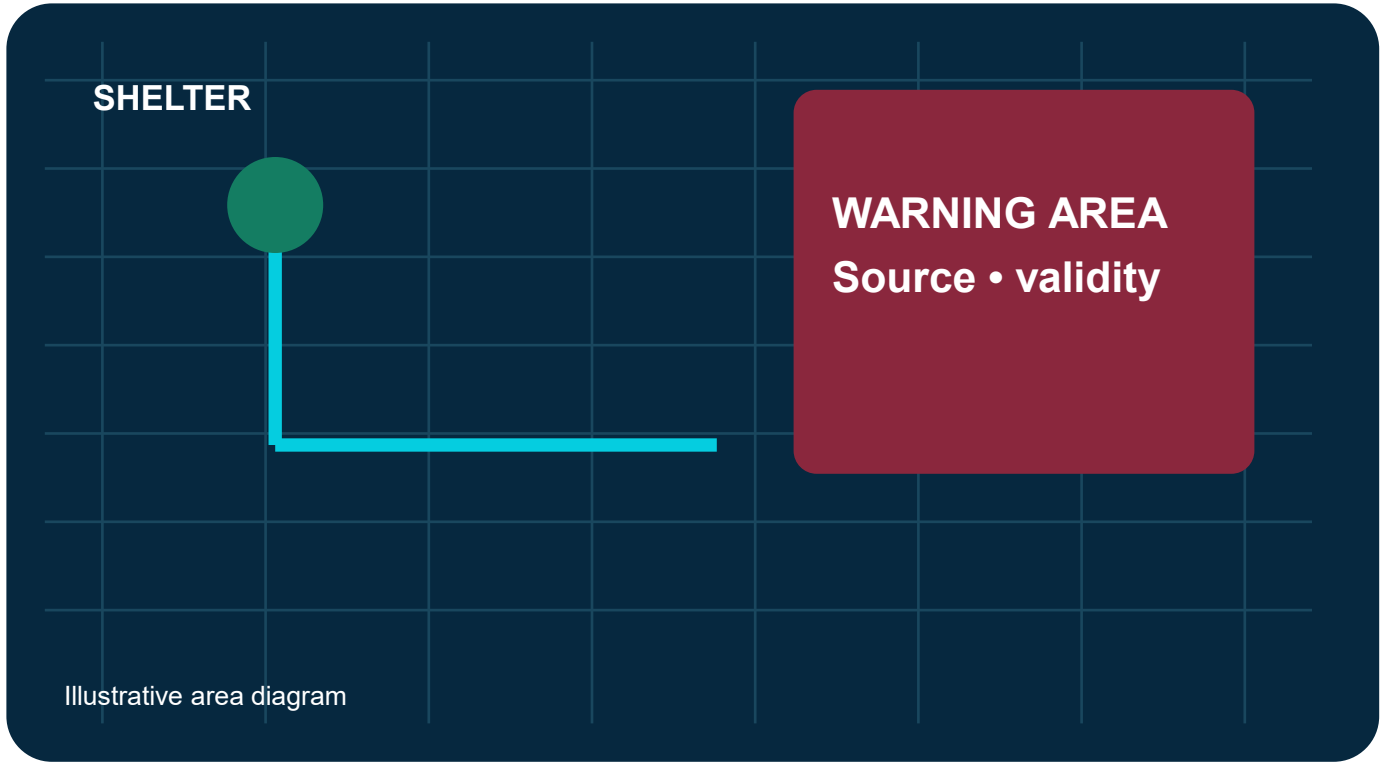
Sent / failed / unknown
Played / interrupted
Human confirmation / escalation

Send updates and cancellations to apps and affected playback devices.

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Area-based warnings lead to local action.

Storms, floods and widespread hazards connect warning sources, shelters and responders.



Receive	Authorized issuer; original, update and cancellation.
Assess	Affected area, people present and assistance needs.
Act	Guidance, shelters, PA and tasks.
Track	Check-ins, contact and expiry updates.

SmartAlarm

Different situations require different responses.

SOS, automatic detection and assistance requests are routed using their context.

FALLS

Wearable / sensor / self-report

Check in → caregiver → assistance if needed

No response prompts a check; it is not a diagnosis.

ROAD ACCIDENTS

App / telematics / external source

Location & road direction → appropriate responders

The people involved and injuries may still be unknown.

DISCREET SOS / ASSISTANCE

SOS button / discreet action

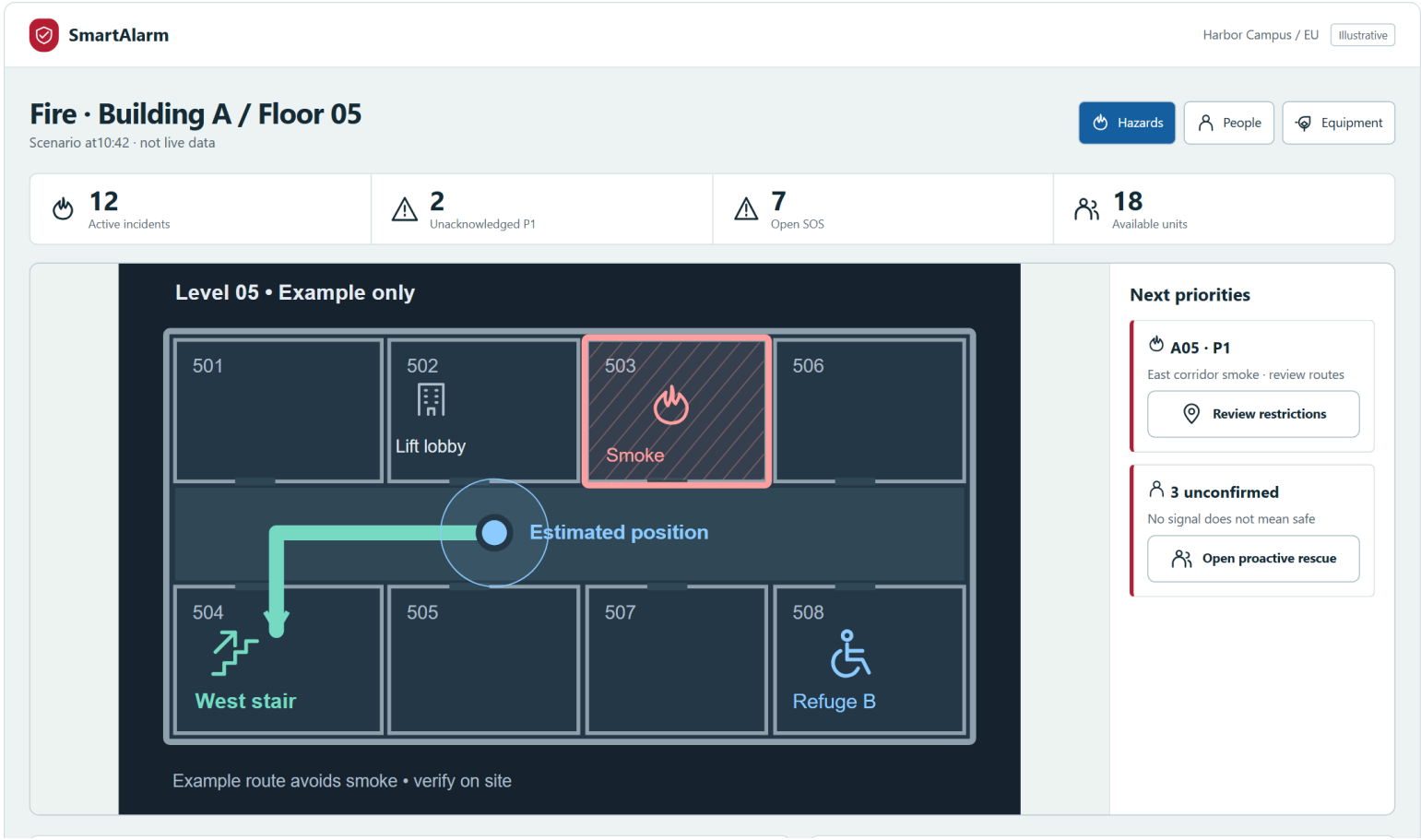
Classify the need → appropriate workflow

Message content and audio must match the sensitivity.

SmartAlarm

An operating picture with evidence and next actions.

Hazard layers • people requiring assistance • dispatch • PA • unresolved information



01 LOCATE
Hazard, person and refuge in one view.

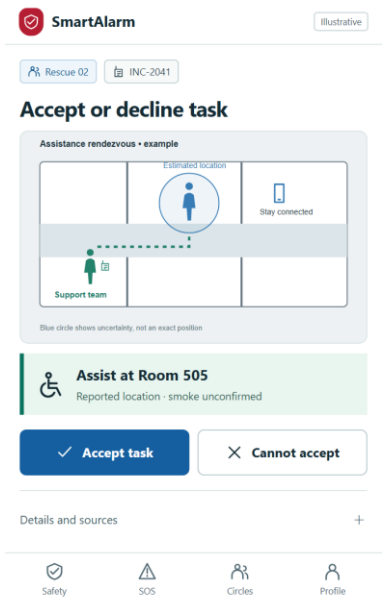
02 PRIORITIZE
Unconfirmed people remain visible.

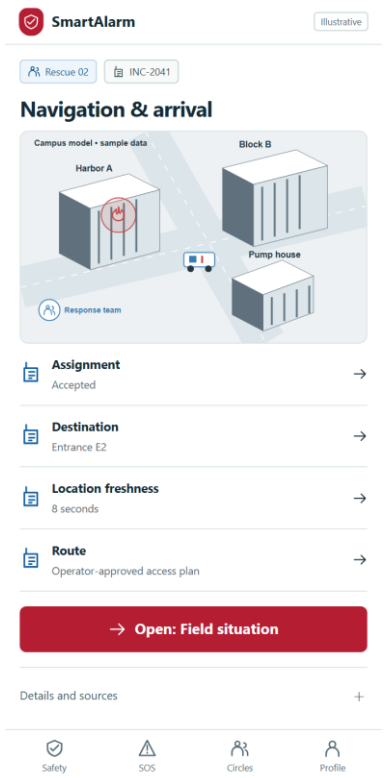
03 COORDINATE
Dispatch and speaker coverage link to evidence.

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A rescue task that can be understood at a glance.

Location • access constraints • team acceptance • field confirmation





ASSIGN → ACCEPT → APPROACH → HAND OVER

A map and short task brief help the responder identify the target and access constraints.

Acceptance is distinct from arrival. Offline updates remain queued until acknowledged.

Functional needs inform assistance; no SOS does not establish safety.

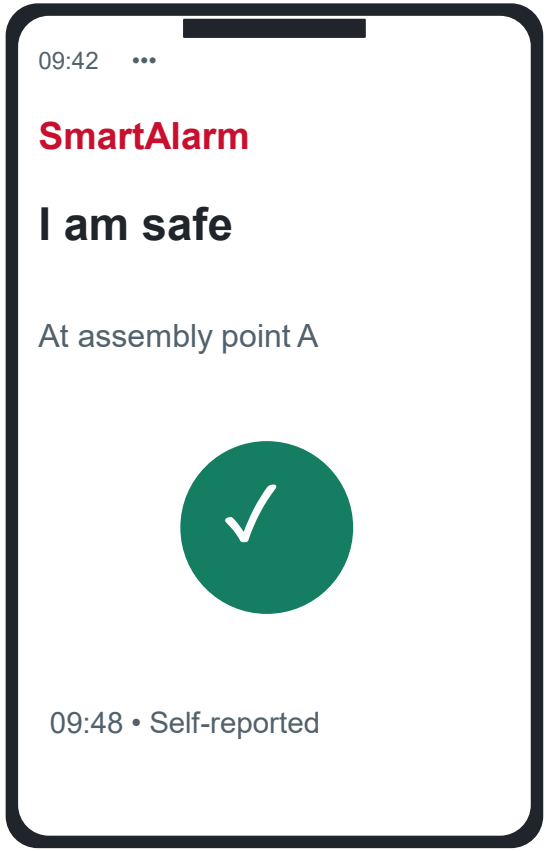
SmartAlarm

A safety update reaches the right recipients.

Relatives can follow status while sensitive information remains protected.



Family receives status authorized for sharing.



Contacts	Authorized to receive updates.
Groups	Family / residents / colleagues.
Operations	Update the person; do not automatically close the incident.

SmartAlarm

AI accelerates understanding, grounded in evidence.

Summarize events, compare procedures and explain options to the responsible decision-maker.

EVIDENCE

Events & timestamps
Location & confidence
Versioned SOPs / maps

ANALYSIS

Summarize changes
Identify missing information
Recommend options & rationale

DECISION

Operator assessment
Workflow authorization
Engine / device execution

If AI is unavailable, alerts, SOS, tested routing and rescue workflows continue to operate.

Two hubs connect the capabilities of SmartAlarm.

Build on FireEscape's Alarm Hub + Integration Hub model for a long-term platform.

PEOPLE	Individuals • family • responders • command • administrators • partners
EXPERIENCES	Smartphone • web command • field app • portals • PA & voice

SMART ALARM HUB

Event → Incident → Warning → Task → Confirmation

Positioning • Routing • Care • Proactive rescue • AI copilot

INTEGRATION HUB

IoT • PA • cameras • drones

Fire panels • pumps • valves

Rescue robots

SHARED PLATFORM Tenant • Access • Data • Event bus • Audit • Observability • Multi-region operations

SmartAlarm

From a raw signal to a unified rescue case.

Extend capabilities through clear domains, data ownership and versioned contracts.

01 EVENT	02 INCIDENT	03 RESPONSE	04 PERSON
Normalize	Correlate	Warn	SOS / assistance
Deduplicate	Verify	Guide	Make contact
Preserve source	Affected area	Assign	Confirm

Shared context	Tenant • site • person • location • time • evidence
Supporting capabilities	Route engine • care/presence • voice • collaboration • AI
Operational assurance	Durable workflows • audit • idempotency • recovery

Add solutions through connectors, with a stable core.

Declare capabilities, adapt protocols and verify behavior before enabling operations.

Declare	Device / source • tenant/site • capability • schema version
Connect	MQTT / HTTP / webhook / gateway • SDK or adapter
Normalize	Identity • timestamp • quality • event and command envelopes
Verify	Dedupe • disconnection • timeout • permissions • compatibility
Operate	Health • credential rotation • replay • revoke • upgrades

PARTNER KIT

API / SDK
Schema examples
Test environment
Conformance test suite
Version management

SmartAlarm

Connect each device through its proven capabilities.

Reading data, issuing warnings and controlling equipment require distinct permissions.

Cameras / sensors	Events, images, connection health and zone.
Wearables / personal devices	Possible falls, SOS and location, subject to sharing permissions.
PA / fire alarms	Status and content playback through the site controller.
Drones / vehicles	Video, telemetry and speakers; authorized pilot and mission.
IOC / partners	Incident records, tasks, status and handover.

CAPABILITY | AUTHORIZATION | EXECUTION | VERIFICATION

CONNECTED FIRE PROTECTION & RESCUE



PHYSICAL FIRE PROTECTION & RESCUE • Fire panels • Pumps, valves & water discharge • Smoke control • Rescue equipment

Connect the water supply, protection systems and rescue response.

Each asset has an identity, a protected area, monitored signals and explicit control rights.

WATER & SUPPRESSION

Tank • pump controller • pump
Valves / waterflow / pressure
Sprinklers • deluge / pre-action

EVACUATION SUPPORT

Fire panel • emergency PA
Smoke control / dampers / doors
Emergency lighting and exit signs

RESCUE EQUIPMENT

Robot / controlled water monitor
Drone / vehicle / speaker
Mission, operator and feedback

Monitoring ≠ Control permission. Command accepted ≠ Physical action completed.

Select interfaces by verified model and permitted function.

Examples checked against manufacturer documentation; installed compatibility still requires acceptance.

Siemens Cerberus PRO	Selected US BACnet gateways: monitoring; regional configurations differ.
NOTIFIER BACNET-GW-3	Supplementary status interface; not a mass-notification trigger path.
Tornatech GPX / GPD	Pump/pressure/fault maps and vendor inputs; verify write rights per firmware.
Potter PFC-4410G3	Local releasing panel and status relays. Ethernet is not proof of a release API.
Bosch PRAESENSA	Documented Open Interface; emergency use depends on configuration/certification.
Shark Robotics Colossus	Robot with water monitor; a public control API has not been established.

Research dossier: 12 device configurations, official sources, vendor RFIs and device-specific tests.

SmartAlarm

A controlled integration path to local controllers.

Normalize data at the edge; dedicated systems retain safety logic and interlocks.

SMARTALARM

Incident / workflow
Authorization & audit
Content / tasks

›

OT DMZ / EDGE

mTLS / outbound
Point–zone mapping
TTL / quality / buffer

›

VENDOR GATEWAY

BACnet / Modbus
OPC UA / vendor API
Approved supervised I/O

›

LOCAL CONTROL

Cause-and-effect
Interlocks / priority
Equipment & feedback

PHYSICAL FEEDBACK • Alarm / trouble • Pressure / waterflow • Valve position • Local/test mode

Loss of cloud must not disable local automatic protection.

Water discharge follows local logic and physical feedback.

Distinguish heat-operated sprinklers from controlled-release zones; there is no universal “on” button.

HEAT-OPERATED SPRINKLERS

Heads respond to local heat.
Pump controllers retain their logic.
SmartAlarm receives alarm / flow / pressure.

DELUGE / CONTROLLED RELEASE

Site-approved releasing controller.
Remote requests only if explicitly permitted.
Interlocks and local override remain active.

REQUEST	VALIDATE	EXECUTE	VERIFY
Zone / purpose	Rights / conditions	Dedicated controller	Flow / pressure

Uncertain outcome → UNKNOWN + reconciliation + accountable operator. No blind release retry.

From manufacturer evidence to accepted site integration.

Standards, product certification and local approval are distinct layers.

Fire / water systems	NFPA 72 / 13 / 20 / 25; EN 54 / ISO 7240 by market and applicable scope.
Interfaces & data	BACnet / Modbus / OPC UA; MQTT / CAP / CloudEvents by use case.
Security & authority	IT/OT zones, scoped IEC 62443 alignment; command rights by model/site.

01 SURVEY	02 FAT	03 SAT	04 HANDOVER
Asset / point list	Simulated faults & commands	Site feedback measurements	Owner / runbook / evidence

SmartAlarm

Scale by region. Sustain the response on site.

Cloud coordinates at scale; edge retains configured and verified local capabilities.

GLOBAL CONTROL PLANE • Tenant directory • Policy • Catalog • Operations

REGIONAL CELL A

Multi-AZ services
Event bus / workflow
Regional data & evidence

REGIONAL CELL B

Data residency boundaries
Scale by tenant / workload
Region-specific DR policy

SITE / EDGE

Gateway • event buffering
Approved maps / content
Local system priority

End-to-end security / Monitoring & audit / Backup–restore / Fault & recovery exercises

Sensitive data is protected by design.

Responders receive the information needed to act; access is limited and auditable.

RIGHT PERSON

Role + organization + site

Current task assignment

Field-level permissions

RIGHT PURPOSE

Separate needs from medical history

Permission-based precise location

Retention and processing basis

AUDITABLE EVIDENCE

Who viewed / changed / commanded

Reason for emergency access

Audit and post-incident review

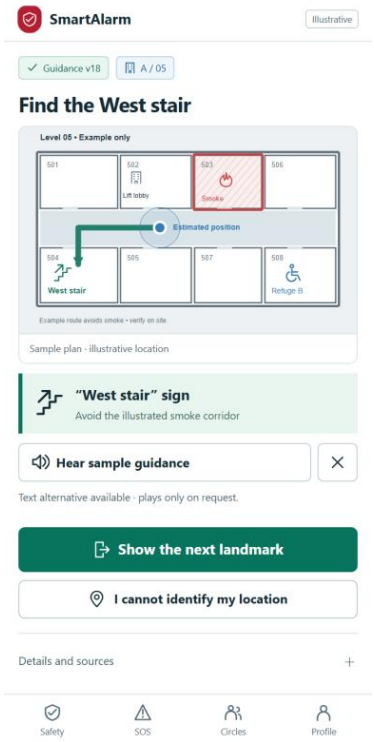
Validate policies for the customer's market and operating model.

Multilingual, from interfaces to rescue messages.

Language, country and time zone are independent settings.



VI



EN



JA

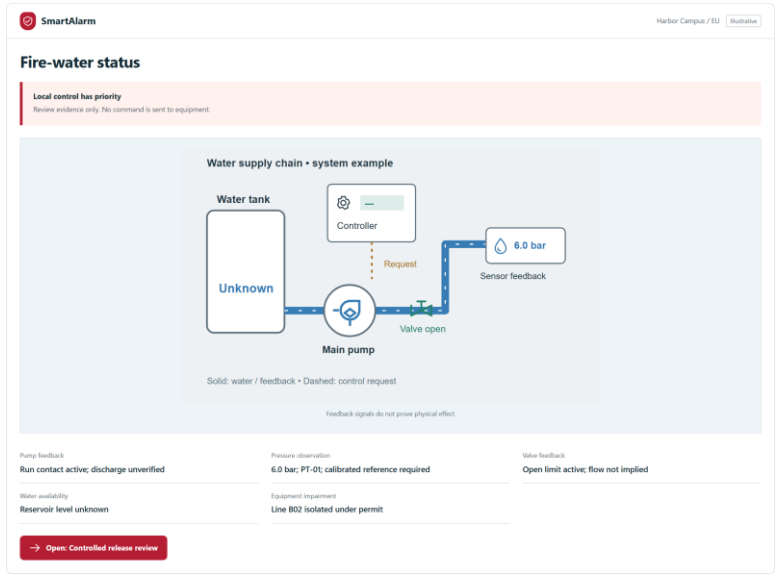
One visual meaning.
Three languages.

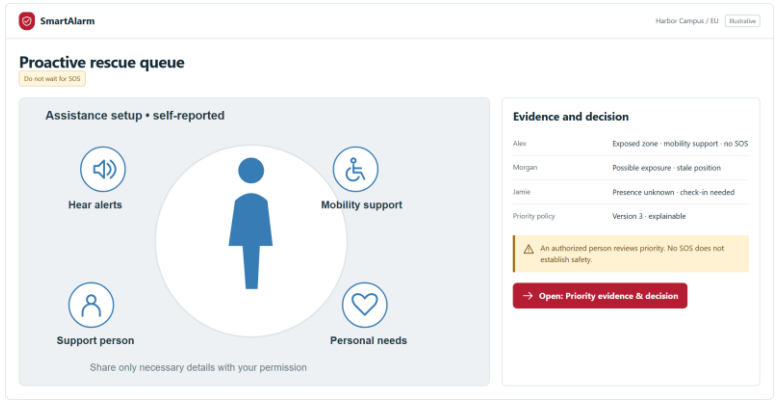
Reviewed emergency content; controlled voice fallback.

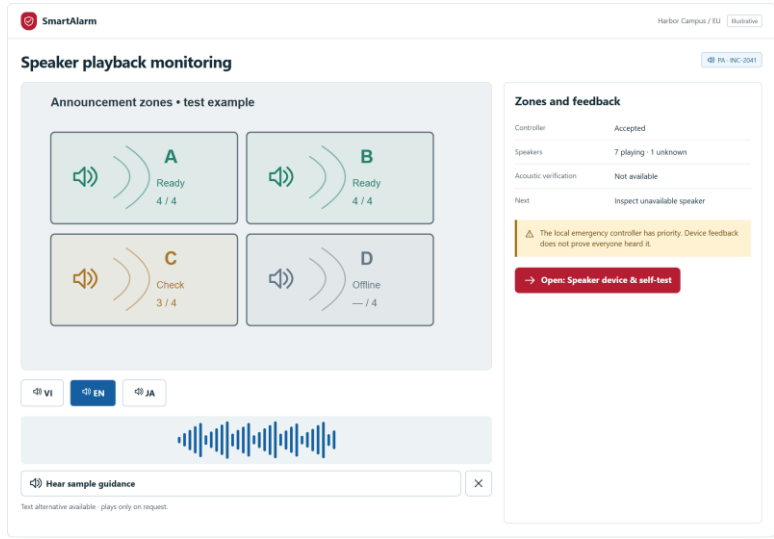
SmartAlarm

241 screens, with visual patterns for each role.

Complete function appendix retained • visual specifications mapped to every screen ID







PHYSICAL SYSTEMS

CITIZEN / FIELD / COMMAND / CONFIGURATION

PROACTIVE RESCUE

SPEAKER COVERAGE

• ALL 241 SCREENS: GALLERY + INTERACTIVE PROTOTYPE

Prove scenarios, beyond individual screens.

Test people, equipment, data, actions and recovery together.

Fire & evacuation	Signal → route/voice → PA → SOS → contact → safety.
Without SOS	Assistance need → presence → check → assignment.
Widespread disaster	Official source → area → update/cancel → shelter.
Network / device failure	Edge, queues, expiry, synchronization and audit.
Access & languages	Tenant isolation, care data, VI/EN/JA, fallback and audio.

SmartAlarm

Keep the full scope. Verify quality at every step.

Acceptance gates turn a long-term architecture into accountable operations.

01 DESIGN	02 BUILD	03 VERIFY	04 OPERATE
Operating model	All domains & roles	End-to-end UAT	Training & handover
Data & permissions	Integrations / locales	Load & faults / DR	Duty teams & support
Equipment survey	Security & automation	On-site exercises	Controlled improvement

Production architecture from day one • Full product scope • Quality gates before go-live

SmartAlarm

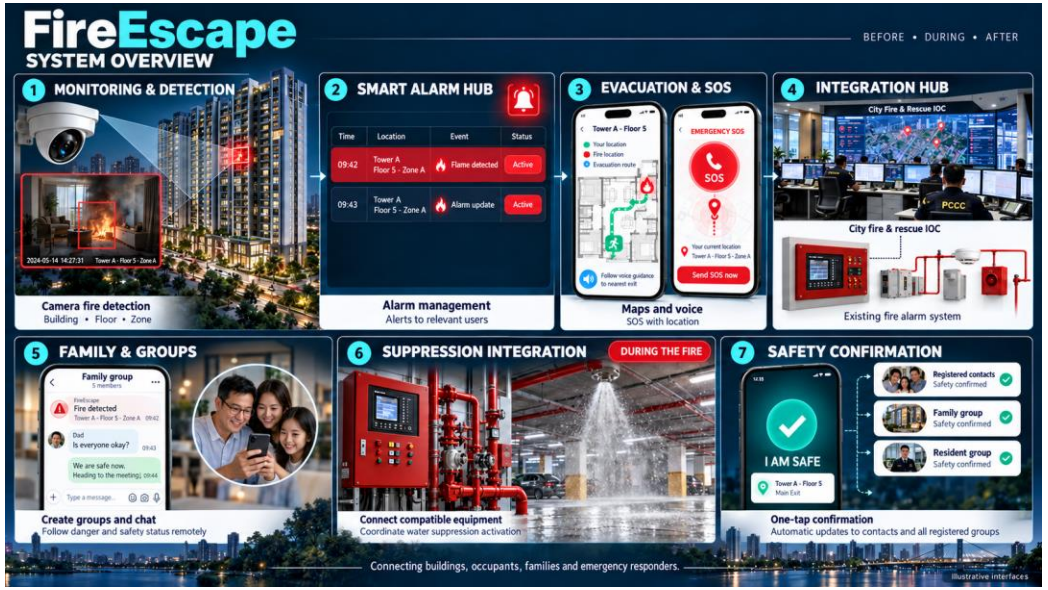
Invest in measurable coordination capability.

Assess value through operational results, with clear scope and responsibilities.

Earlier awareness	Measure signal-to-verification time by incident type.
Clearer contact status	Measure task acceptance, contact and unresolved cases.
Consistent coordination	Measure message consistency, handovers and exercise outcomes.
Easier expansion	Measure integration effort, version compatibility and health.
Agree before deployment	Sites/users/devices, data regions, channels, responders and SLA.

From FireEscape to SmartAlarm: a grounded expansion.

Preserve the existing response cycle and add the capabilities requested by the product owner.



FIRESCAPE • ORIGINAL MATERIAL

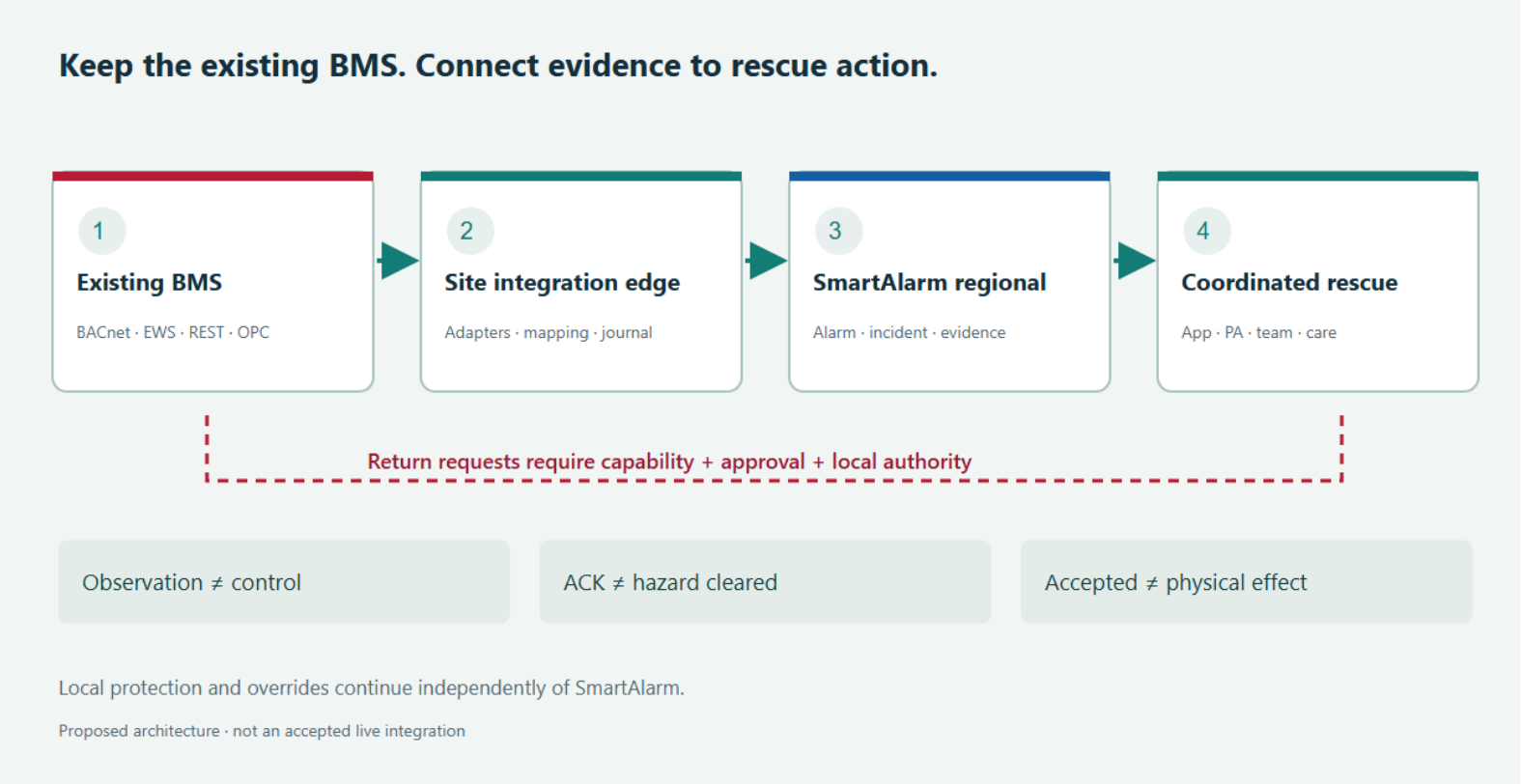
Retain	Cameras, floors/zones, maps/voice, SOS, hubs, groups and safety updates.
Extend	Multiple hazards, care/proactive rescue, PA/drones, positioning/routing and AI.
Long-term foundation	Multiple tenants/industries/regions; VI/EN/JA; security and operations.

Images and interfaces illustrate the proposed solution. This deck does not claim the extensions are already deployed.

SmartAlarm

Keep the existing BMS. Connect it to rescue action.

Building information becomes evidence for command, guidance and human assistance.



UNDERSTAND

Alarms, assets, floors and zones.

COORDINATE

App, PA, teams and people needing help.

CONTROL

Authorized requests; source-backed feedback.

SmartAlarm

One connector platform. Product-specific profiles.

8 vendor/product families • 11 profiles • version, licence and capability qualification.

Siemens
Desigo CC / WSI

Schneider Electric
EBO / EWS + SmartConnector

Johnson Controls
Metasys / REST + SSE

Honeywell
EBI / WEBs — separate profiles

Tridium
Niagara local / NDS cloud

Delta Controls
enteliWEB / API entitlement

Automated Logic
WebCTRL events / SOAP + SDK

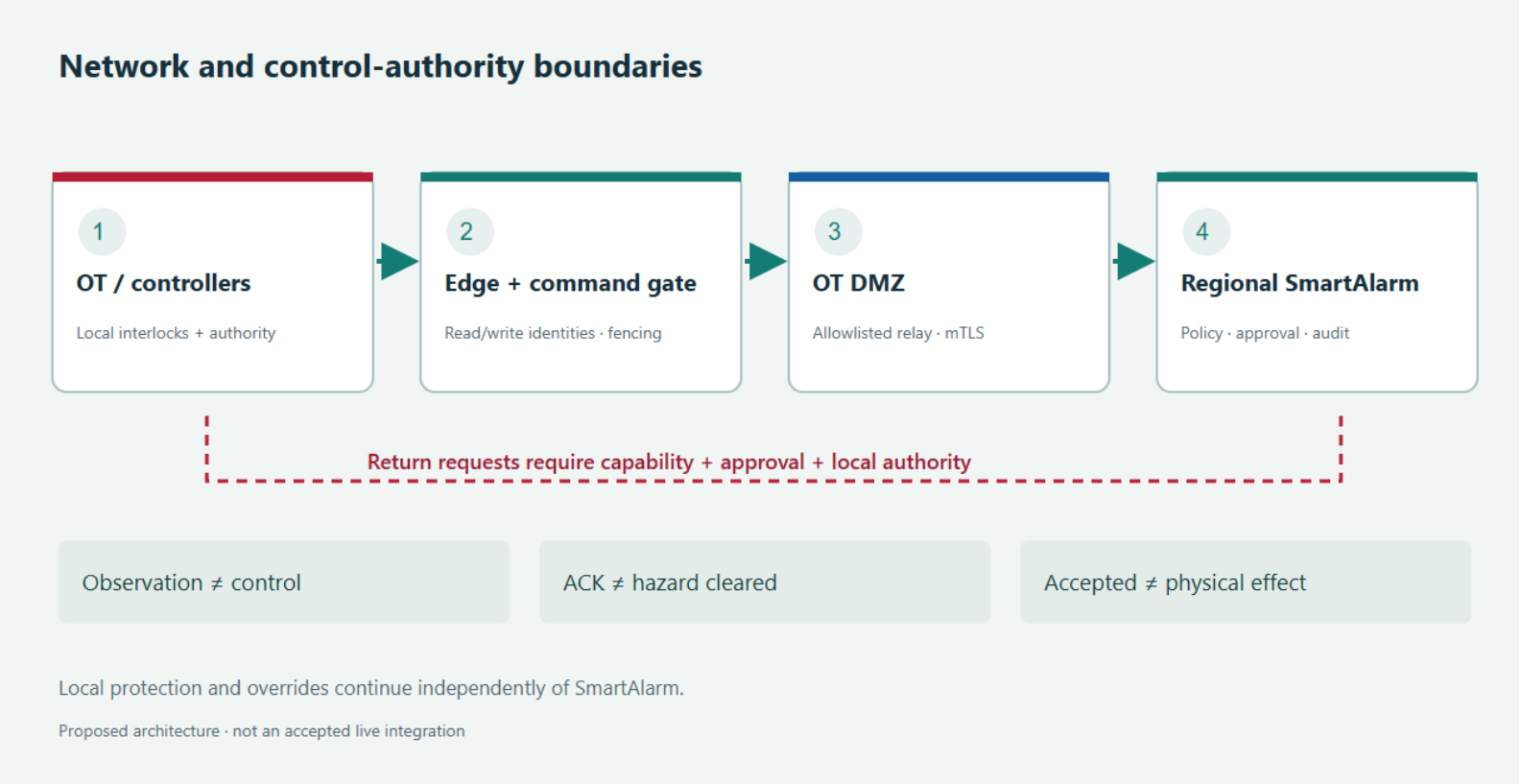
Distech Controls
EC-Net / Niagara OEM

A shared protocol does not establish compatibility for every operation or product version.

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Separate observation from governed control.

Site gateway • OT/DMZ boundary • normalized data • regional SmartAlarm services.

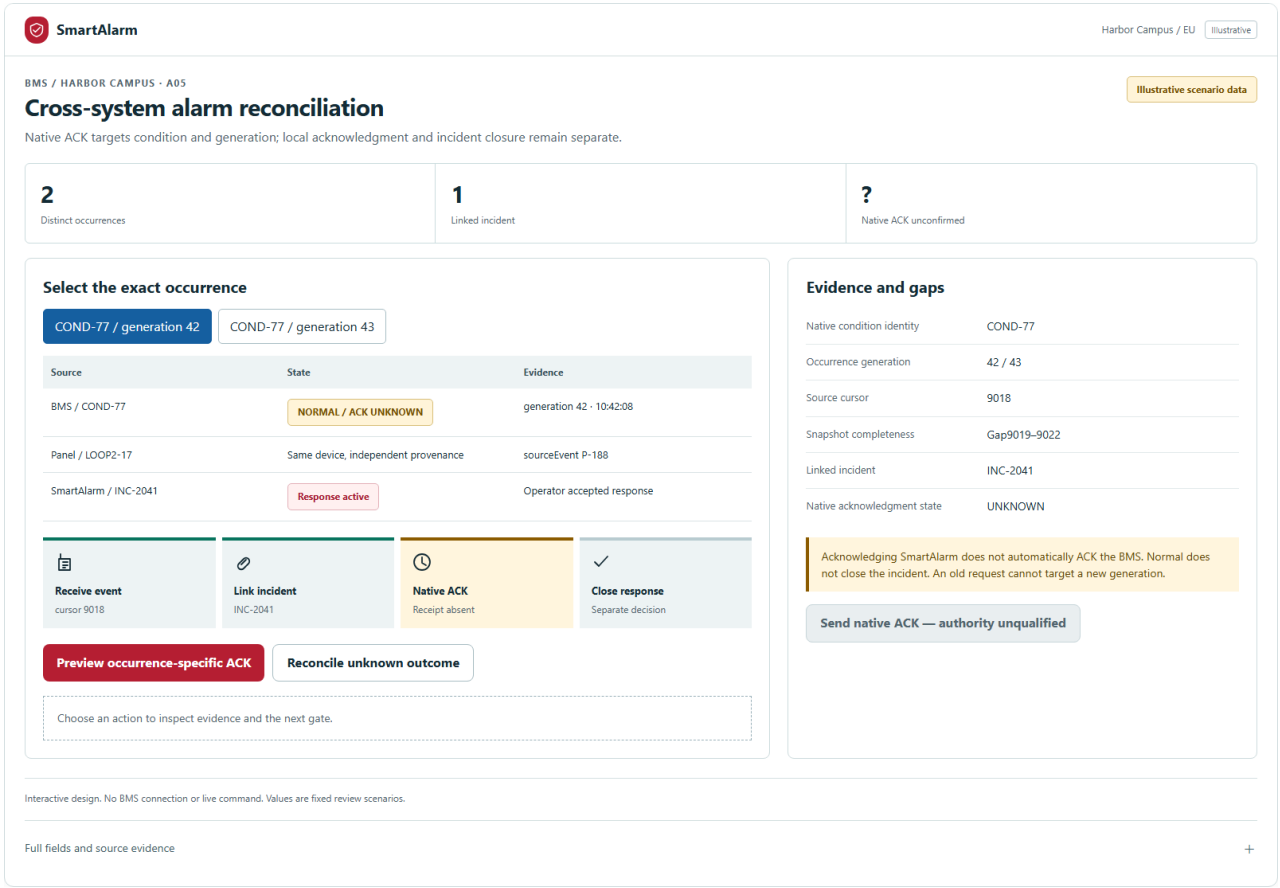


Local BMS/controller authority remains. LLMs do not issue direct commands. Link loss does not prove relinquishment or safety.

SmartAlarm

One incident record, with multiple evidence sources.

Reconcile BMS and direct-panel feeds; retain source, timestamps, quality and acknowledgement history.



01 Receive and check freshness

02 Correlate without erasing provenance

03 Assess routes and people at risk

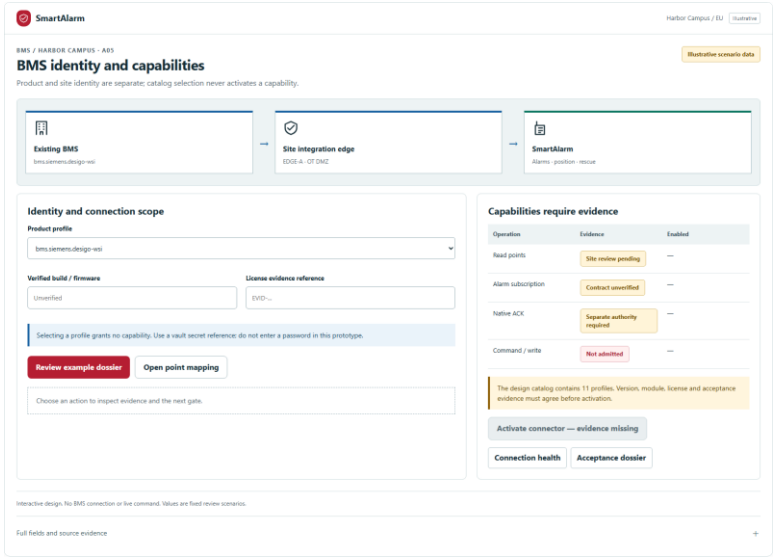
04 Coordinate app / PA / teams

05 Separate rescue confirmation from ACK

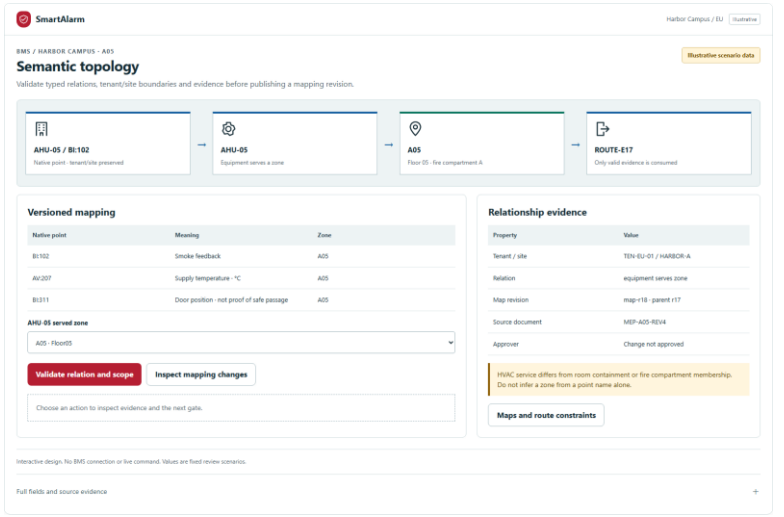
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From connection to mapping and evidence-based qualification.

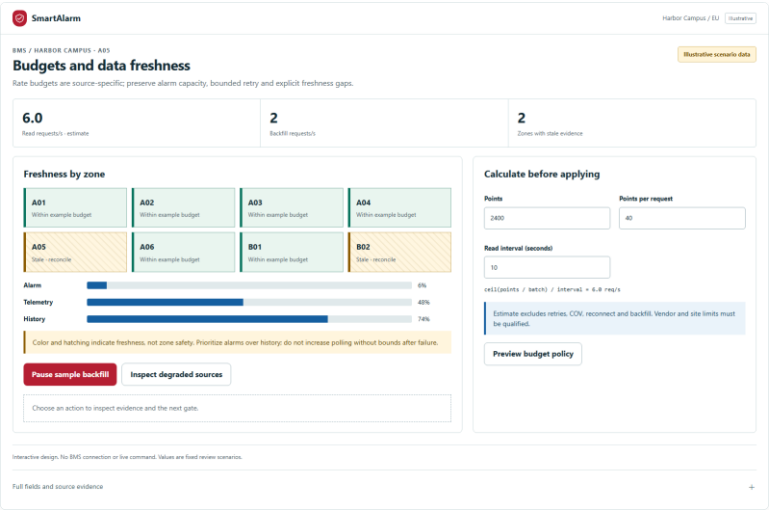
Versioned configuration; qualify actual capabilities before enabling permissions.



1 · Connection inventory



2 · Mapping and preview

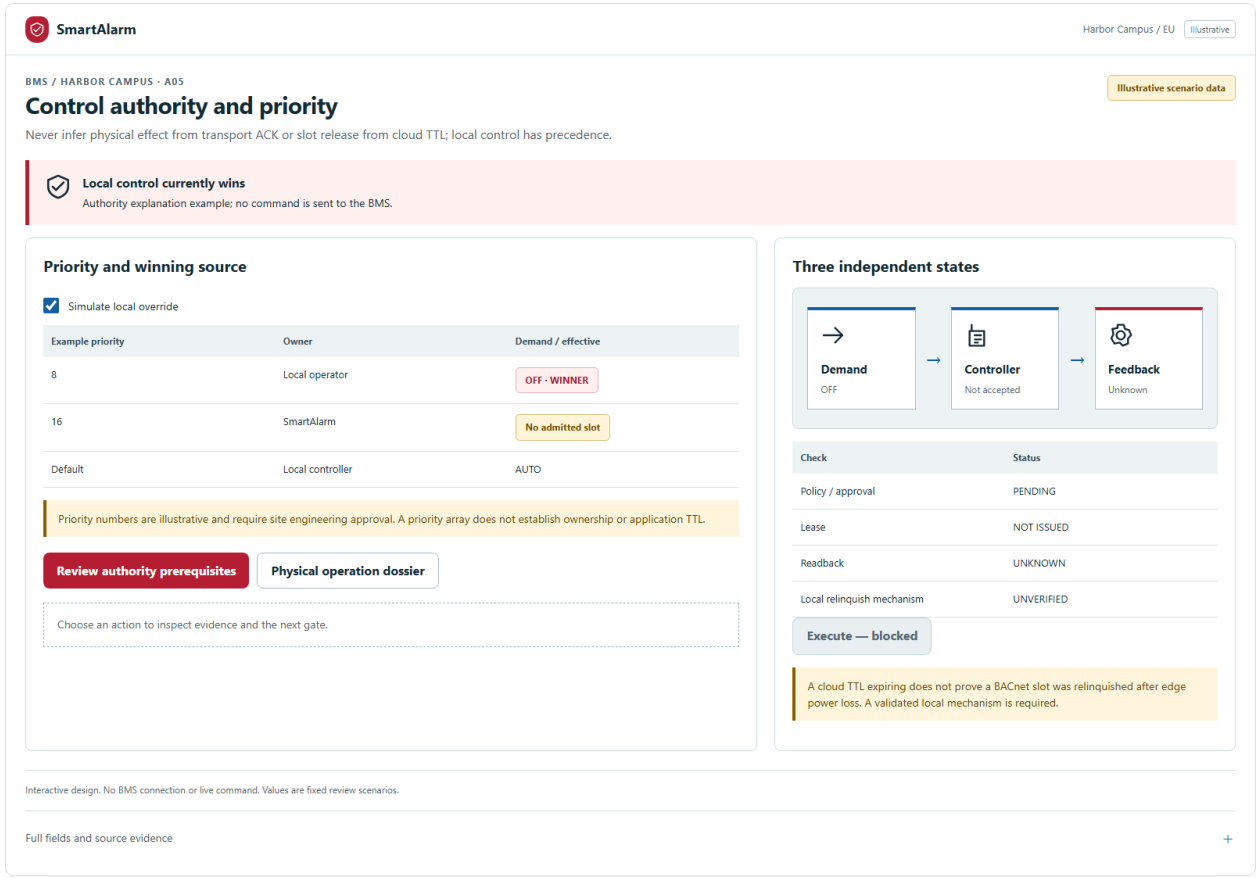


3 · Qualification / FAT / SAT

SmartAlarm

Receipt, execution and physical effect are separate states.

Preview impact → approval → authorized gateway → reconcile feedback.



LOCAL AUTHORITY

Overrides and interlocks take priority.

UNKNOWN OUTCOME

Reconcile; never blindly retry.

COMPLETE SCOPE

40 FR-BMS + 8 new screens.

AIPOWER
Your AI Power Partner

SmartAlarm

Detect. Guide. Rescue. Confirm.

Connect equipment with people —
and recognize those who need help, even without an SOS.

CONNECTED SAFETY & RESCUE / INTERNATIONAL / VI • EN • JA

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The complete baseline, with traceable IDs.

Beyond primary screens: business workflows, detailed functions and background processing.

37

BUSINESS WORKFLOWS

241

SCREEN-LEVEL FR FUNCTIONS

312

BACKGROUND / SYSTEM REQUIREMENTS

A · End-to-end workflows

B · Every detailed function, organized by domain

C · Background processing, integration, controls and quality requirements

SmartAlarm

Cross-system business workflows | 1/7

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
WF-01	Provision tenant/region, SSO/MFA, invitations and workspace access without account enumeration.	14 screens
WF-02	Manage site/floor/zone, approved plan revisions and location quality.	6 screens
WF-03	Durably ingest multi-source events with source/tenant validation, deduplication and quarantine.	5 screens
WF-04	Review evidence and create/link/split incidents while retaining audit and ownership.	7 screens
WF-05	Submit need-specific SOS with manual location, offline pending and reviewed cancellation.	10 screens

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Cross-system business workflows | 2/7

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
WF-06	Check possible falls with a server timer and escalate non-response without diagnosis.	2 screens
WF-07	Report and dispatch road incidents using location, road direction and unknown injury status.	1 screens
WF-08	Process CAP/disaster advisories with issuer, mode, polygon, lineage and expiry validation.	4 screens
WF-09	Author and approve exact warning content, audience and locale versions.	4 screens
WF-10	Track per-channel delivery, authenticated callbacks, bounded retries/fallback and escalation.	6 screens

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Cross-system business workflows | 3/7

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
WF-11	Dispatch capable teams, accept tasks atomically, update progress and hand over ownership.	17 screens
WF-12	Record sourced personal status and share with circles/contacts under consent.	12 screens
WF-13	Maintain shelters and approved guidance with versioned, expiring offline packs.	9 screens
WF-14	Enroll devices, manage certificates/health, maintain/calibrate and expose coverage gaps.	5 screens
WF-15	Control devices only within capability/interlock/TTL rules and verified outcome evidence.	2 screens

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Cross-system business workflows | 4/7

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
WF-16	Plan/review drone missions with pilot authority, telemetry and local lost-link policy.	5 screens
WF-17	Manage connector lifecycle from manifest/sandbox to activation, rotation and deprecation.	14 screens
WF-18	Provide locale/timezone support, residency and verified site/country emergency contacts.	9 screens
WF-19	Apply retention/legal holds, scoped export/deletion and immutable audit.	11 screens
WF-20	Close incidents, analyze/improve response, isolate drills and acknowledge shift handovers.	30 screens

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Cross-system business workflows | 5/7

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
WF-21	Reach people without phones through approved PA zones/content, playback, live speech and fallback.	9 screens
WF-22	Guide step-by-step using graph/hazard/accessibility constraints, replanning and SOS fallback.	10 screens
WF-23	Fuse uncertain position evidence and distinguish observed presence from registration.	4 screens
WF-24	Optimize feasible routes under hard constraints, capacity and cohort balancing.	3 screens
WF-25	Ground AI summaries/SOP checks/drafts in evidence; human review, no autonomous actuation.	4 screens

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Cross-system business workflows | 6/7

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
WF-26	Manage care profiles, caregiver authority, emergency subsets and audited access.	7 screens
WF-27	Proactively identify/prioritize assistance before SOS using presence and evidence.	3 screens
WF-28	Bind model/firmware/capability, point/zone mappings and approved cause-and-effect.	6 screens
WF-29	Supervise physical signals with provenance/freshness and independent local protection.	3 screens
WF-30	Authorize/execute permitted commands, reconcile UNKNOWN and support local takeover.	5 screens

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Cross-system business workflows | 7/7

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
WF-31	Isolate impairment/tests, assign compensating measures and verify restoration.	2 screens
WF-32	Maintain compatibility/certification, FAT/SAT and configuration-specific capability gates.	3 screens
WF-33	Dispatch robot/water-monitor missions with operators, limits and verified outcomes.	1 screens

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Access | 1/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-A01	Welcome & region: Choose your safety region; continue.	WF-01
FR-A02	Sign in: Your workspace, securely connected; sign in.	WF-01
FR-A03	Verify identity: Confirm this sign-in; verify.	WF-01
FR-A04	Workspace switcher: Choose an organization; enter workspace.	WF-01
FR-A05	Recover access: Restore access without exposing account existence; send recovery link.	WF-01

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Access | 2/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-A06	Accept invitation: Review your organization invitation; accept invitation.	WF-01
FR-A07	Permissions & consent: Choose how SmartAlarm supports you; save choices.	WF-01
FR-A08	Session & device security: Review devices with access; review session.	WF-01
FR-A09	Revoke session: End access on the selected device; revoke selected session.	WF-01
FR-A10	Profile & accessibility: Make the interface work for you; save profile.	WF-01

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Citizen | 1/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-C01	Safety home: A clear next step, when it matters; view active alert.	WF-13
FR-C02	Alert detail: Fire reported in Building A; view guidance.	WF-13
FR-C03	Evacuation guidance: Use the published evacuation plan; i need assistance.	WF-13
FR-C04	Voice & accessible guidance: Read or listen to the same instruction; return to plan.	WF-13
FR-C05	SOS type: What kind of help do you need?; continue with selected need.	WF-05

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Citizen | 2/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-C06	SOS location & details: Tell responders where to find you; review SOS.	WF-05
FR-C07	Confirm SOS: Send this request to campus command?; send SOS.	WF-05
FR-C08	SOS tracking: Your request has been received; contact assigned team.	WF-05
FR-C09	Incident conversation: Stay connected to the response team; send message.	WF-05
FR-C10	Update or cancel SOS: Update your situation; submit update.	WF-05

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Citizen | 3/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-C11	Confirm personal safety: Let your circle know your status; confirm my safety.	WF-12
FR-C12	Safety receipt: Your safety update was saved; view delivery details.	WF-12
FR-C13	Notification inbox: Updates that need your attention; open alert.	WF-10
FR-C14	Nearby hazards: Know what is happening nearby; view hazard.	WF-08
FR-C15	Disaster advisory: Flood watch for Riverside district; view shelter information.	WF-08

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Citizen | 4/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-C16	Shelters & assembly points: Find a designated assistance point; open shelter details.	WF-13
FR-C17	Shelter details: North shelter; save shelter offline.	WF-13
FR-C18	Offline safety pack: Essential information on your device; review cached plan.	WF-13
FR-C19	My circles: The people who should hear from you; open family.	WF-12
FR-C20	Circle conversation: Family; view member status.	WF-12

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Citizen | 5/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-C21	Create or edit circle: Choose members and sharing rules; save circle.	WF-12
FR-C22	Circle member status: A status is a report, not a guarantee; send check-in request.	WF-12
FR-C23	Emergency contacts: Who should receive your updates?; save contacts.	WF-12
FR-C24	Care & fall detection: Configure a supported fall detector; review test alert.	WF-06
FR-C25	Possible fall check-in: A possible fall was detected; i need help now.	WF-06

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Citizen | 6/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-C26	Road incident report: Report a road incident safely; review SOS.	WF-07
FR-C27	Discreet assistance: Request help with a quiet interface; send discreet request.	WF-05
FR-C28	My reports & history: Your previous requests; view request.	WF-05
FR-C29	Privacy & data requests: Control optional sharing; review data request.	WF-19
FR-C30	Data request receipt: Request DR-202 received; back to privacy.	WF-19

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Citizen | 7/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-C31	Circle membership & invitations: Control who receives your updates; manage invitation.	WF-12
FR-C32	Invite circle member: Invite someone you trust; create invitation.	WF-12
FR-C33	Report or block conversation: Control unwanted social contact; submit report.	WF-12
FR-C34	Notification preferences: Choose channels and quiet hours; save preferences.	WF-10
FR-C35	Voice evacuation navigation: Follow reviewed steps away from the danger area; show next instruction.	WF-22

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Citizen | 8/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-C36	Navigation step & orientation: Use the WEST STAIR sign as your landmark; repeat voice instruction.	WF-22
FR-C37	Route change notification: The previous route is no longer recommended; review updated route.	WF-22
FR-C38	Position uncertain / no route: We cannot verify a route from your position; request assistance.	WF-22
FR-C39	Outdoor hazard avoidance: Leave the affected area using reviewed guidance; view next route step.	WF-22
FR-C40	Voice language & guidance settings: Choose how guidance reaches you; save guidance preferences.	WF-22

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Responder | 1/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-R01	Shift & readiness: Ready for your next assignment; view assignments.	WF-11
FR-R02	Assignment inbox: Assignments for Rescue 02; open assignment.	WF-11
FR-R03	Accept or decline task: Assist person at Room 505; accept assignment.	WF-11
FR-R04	Navigation & arrival: Building A · East access; report arrival.	WF-11
FR-R05	Field situation: Floor 05 response context; open assistance record.	WF-11

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Responder | 2/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-R06	Assistance & triage record: Record observations, not diagnosis; save observation.	WF-11
FR-R07	Task status & handover: Hand over with a clear record; submit handover.	WF-11
FR-R08	Task completion: Assignment completed; back to shift.	WF-11
FR-R09	Team communications: Incident channel · INC-2041; send message.	WF-11
FR-R10	Evidence capture: Attach a field observation; queue attachment.	WF-11

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Responder | 3/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-R11	Offline outbox: Updates waiting to sync; resolve sync conflict.	WF-11
FR-R12	Resolve field conflict: A newer assignment state exists; keep note and refresh.	WF-11

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Command | 1/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-O01	Command overview: Every signal. One coordinated response; open priority incident.	WF-20
FR-O02	Live situation map: See incidents across your authorized region; open incident.	WF-20
FR-O03	Signal triage queue: Separate evidence from assumptions; inspect signal.	WF-03
FR-O04	Incident workspace: Fire · Building A / Floor 05; open dispatch.	WF-20
FR-O05	Signal evidence review: Camera and sensor evidence; create incident from signal.	WF-04

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Command | 2/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-O06	Create incident: Establish the incident record; create incident.	WF-04
FR-O07	Correlate or split incidents: Review possible duplicate reports; save correlation.	WF-04
FR-O08	Dispatch board: Match the task to a capable team; review assignment.	WF-11
FR-O09	Assign responder: Confirm ownership of the task; assign team.	WF-11
FR-O10	Response timeline: A shared record of decisions; review people status.	WF-11

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Command | 3/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-O11	People accountability: Know what is reported and what is unknown; open person.	WF-12
FR-O12	Person assistance detail: Alex Morgan · SOS-041; assign assistance.	WF-12
FR-O13	Compose warning: Prepare a targeted warning; preview audience.	WF-09
FR-O14	Warning preview & audience: Review message and recipients; submit for approval.	WF-09
FR-O15	Approve warning: Independent approval before broadcast; approve warning.	WF-09

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Command | 4/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-O16	Delivery & escalation: Sending does not mean received; review failed deliveries.	WF-10
FR-O17	Delivery failure review: Resolve notification gaps; queue fallback.	WF-10
FR-O18	Physical command approval: Review a proposed equipment action; approve command.	WF-15
FR-O19	Physical command status: Track controller evidence; refresh command evidence.	WF-15
FR-O20	Incident closure review: Resolve outstanding obligations; close incident.	WF-20

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Command | 5/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-O21	After-action review: Learn from the response; create improvement action.	WF-20
FR-O22	Improvement action: Give follow-up work an owner; save action.	WF-20
FR-O23	Drill scheduler: Practice without contacting live recipients; open drill control.	WF-20
FR-O24	Drill control room: EXERCISE · Building A evacuation; review drill report.	WF-20
FR-O25	Shift handover: Transfer active operational responsibility; record handover.	WF-20

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Command | 6/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-O26	Shelter operations: Publish fresh shelter information; edit shelter.	WF-13
FR-O27	Edit shelter status: Update a designated shelter; publish update.	WF-13
FR-O28	Regional disaster feed: Review trusted public advisories; inspect advisory.	WF-08
FR-O29	CAP advisory review: Preserve the issuing authority's context; create local response.	WF-08
FR-O30	Search & audit timeline: Find an event without changing history; inspect audit entry.	WF-19

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Command | 7/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-O31	Incident archive & reopen: Review closed incidents with full history; review reopening.	WF-20
FR-O32	Reopen incident: Record why response needs to resume; reopen incident.	WF-20
FR-O33	Federated incident sharing: Share a controlled incident projection; review partner exchange.	WF-17
FR-O34	Accessibility assistance queue: Assign support based on reported needs; assign capable unit.	WF-11
FR-O35	Hazard zones & route restrictions: Publish time-bounded movement restrictions; review guidance impact.	WF-22

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Command | 8/8

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-O36	Guidance distribution & receipts: Keep app and speaker instructions consistent; open speaker operations.	WF-22

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Sites | 1/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-S01	Site directory: One location model across industries; open site.	WF-02
FR-S02	Site structure: Harbor Campus / Building A; edit location.	WF-02
FR-S03	Location editor: Link physical and geographic context; save location.	WF-02
FR-S04	Floor plan editor: Publish a reviewed evacuation map; review plan publication.	WF-02
FR-S05	Plan approval & versions: Keep published guidance traceable; publish reviewed draft.	WF-02

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Sites | 2/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-S06	Device inventory: Know which devices can be trusted; open device.	WF-14
FR-S07	Device detail & health: CAM-A05-03; review device configuration.	WF-14
FR-S08	Provision device: Bind a device to an authorized tenant; enroll device.	WF-14
FR-S09	Camera wall: Review selected camera sources; open detection evidence.	WF-14
FR-S10	Maintenance & calibration: Keep suppression visible during maintenance; schedule maintenance.	WF-14

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Sites | 3/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-S11	Rule library: Detection policies by hazard and site; edit rule.	WF-04
FR-S12	Rule editor & simulation: Test a rule against recorded events; view simulation results.	WF-04
FR-S13	Rule test & approval: Review behavioral changes before release; return rule for changes.	WF-04
FR-S14	AI model monitoring: Track detector behavior by site; open review queue.	WF-04
FR-S15	Evacuation graph editor: Build a versioned indoor and outdoor route graph; review route validation.	WF-22

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Sites | 4/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-S16	Route simulation & approval: Test hazards, accessibility and uncertainty; publish reviewed graph.	WF-22

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Integrations | 1/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-I01	Connector catalog: Connect systems through declared capabilities; configure connector.	WF-17
FR-I02	Connector setup: Create a scoped connection; test connection.	WF-17
FR-I03	Schema mapping: Map source data into a versioned event; run contract tests.	WF-03
FR-I04	Connector test console: Prove behavior before enabling traffic; enable connector.	WF-17
FR-I05	Connector health: Monitor transport and processing separately; inspect dead letters.	WF-17

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Integrations | 2/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-I06	Dead-letter & replay: Replay only after inspecting the failure; replay event.	WF-17
FR-I07	API clients & scopes: Grant only required API access; save client.	WF-17
FR-I08	Webhook subscriptions: Deliver filtered incident updates; send test payload.	WF-17
FR-I09	Partner exchange agreements: Define cross-organization sharing; save agreement draft.	WF-17
FR-I10	IOC exchange history: Verify what a partner actually received; inspect payload.	WF-17

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Integrations | 3/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-I11	Drone fleet: Mission readiness at a glance; plan mission.	WF-16
FR-I12	Drone mission planning: Observe the perimeter of Building A; review mission approval.	WF-16
FR-I13	Drone preflight & approval: Local authority remains with the pilot; approve mission.	WF-16
FR-I14	Drone mission telemetry: DR-01 · observation mission; review mission record.	WF-16
FR-I15	Drone mission record: Preserve mission evidence; return to fleet.	WF-16

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Administration | 1/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-T01	Tenant overview: Operate across regions with clear boundaries; manage organization.	WF-20
FR-T02	Organization & regions: Choose residency and local defaults; save settings.	WF-18
FR-T03	Users & invitations: Review access across the organization; edit membership.	WF-01
FR-T04	Role & scope editor: Separate operational duties; save role.	WF-01
FR-T05	On-call & escalation: Escalate when acknowledgment is missing; save escalation.	WF-10

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Administration | 2/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-T06	Message templates & languages: Publish reviewed instructions per locale; preview template.	WF-09
FR-T07	Hazard taxonomy & playbooks: Extend categories without changing the core; save draft playbook.	WF-20
FR-T08	Playbook approval: Approve the operational sequence; publish playbook.	WF-20
FR-T09	Retention & legal holds: Apply policies by data class and region; save policy draft.	WF-19
FR-T10	Privacy request review: Review request DR-202; approve export.	WF-19

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Administration | 3/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-T11	Audit record: Immutable decision evidence; return to audit search.	WF-19
FR-T12	Service health & recovery: Separate degraded service from silent failure; open recovery runbook.	WF-20
FR-T13	Recovery runbook: Regional incident data recovery; record drill outcome.	WF-20
FR-T14	Usage & subscription: Capacity and commercial visibility; review capacity policy.	WF-20
FR-T15	Capacity & quota policy: Protect emergency traffic during bursts; save capacity draft.	WF-20

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Administration | 4/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-T16	Analytics & export: Measure response without hiding gaps; prepare report.	WF-20
FR-T17	Report schedule: Deliver an authorized operational report; save schedule draft.	WF-20

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

System states | 1/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-X01	No connection / SOS pending: Your SOS has not reached the server; view offline outbox.	WF-05
FR-X02	Permission unavailable: Location permission is off; enter location manually.	WF-05
FR-X03	No active incidents: No active incidents in this view; review device health.	WF-20
FR-X04	Access denied: This record is outside your access scope; choose workspace.	WF-01
FR-X05	Service unavailable: This service is temporarily unavailable; open offline pack.	WF-20

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

System states | 2/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-X06	Unsaved changes & conflict: A newer version was published; review latest version.	WF-02
FR-X07	Empty search & filter reset: No matches for these filters; reset search.	WF-20
FR-X08	Expired session: Sign in again to continue; sign in.	WF-01
FR-X09	Loading & stale data: Refreshing the situation view; return to command.	WF-20
FR-X10	Attachment rejected: This attachment could not be accepted; choose another attachment.	WF-11

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Platform | 1/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-G01	Regional operations: Production cells, independently operated; inspect regional cell.	WF-20
FR-G02	Regional cell detail: EU cell · operational readiness; review recovery readiness.	WF-20
FR-G03	Disaster recovery drill: Validate recovery with recorded evidence; record drill evidence.	WF-20
FR-G04	Recovery verification: Verify data and ownership before routing; review regional operations.	WF-20
FR-G05	Tenant provisioning: Provision a production tenant boundary; review provisioning plan.	WF-18

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Platform | 2/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-G06	Tenant lifecycle: Northstar · lifecycle controls; request support access.	WF-18
FR-G07	Time-bound support access: Request the smallest support scope; submit access request.	WF-19
FR-G08	Support access approval: Approve a scoped support session; approve scoped access.	WF-19
FR-G09	Release & schema compatibility: Release only compatible contracts; review release gate.	WF-20
FR-G10	Release gate & rollback: Verify behavior before promotion; record release decision.	WF-20

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Platform | 3/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-G11	Security incidents: Respond to platform threats; open security case.	WF-19
FR-G12	Security case & containment: Contain the source without hiding coverage gaps; record containment.	WF-19

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Developer | 1/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-D01	Developer workspace: Build against explicit contracts; open API explorer.	WF-17
FR-D02	API explorer: POST /v1/events; run sandbox request.	WF-03
FR-D03	Sandbox request results: Inspect the complete receipt; open schema registry.	WF-03
FR-D04	Schema registry: Track compatibility before upgrading; view schema contract.	WF-03
FR-D05	Connector manifest editor: Declare capabilities explicitly; submit for conformance.	WF-17

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Developer | 2/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-D06	Conformance certification: Review adapter behavior under failure; approve adapter version.	WF-17
FR-D07	Webhook delivery trace: Understand every delivery attempt; review subscription.	WF-10
FR-D08	Credential rotation: Rotate credentials without exposing secrets; create rotation request.	WF-17

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Localization | 1/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-L01	Language coverage: Three languages, one product meaning; open translation workspace.	WF-18
FR-L02	Translation editor: Translate a versioned message; submit translation review.	WF-18
FR-L03	Translation review: Review terminology and emergency meaning; approve translation version.	WF-18
FR-L04	Voice & pronunciation: Keep speech aligned with reviewed text; save voice configuration.	WF-18
FR-L05	Locale fallback policy: Make missing translations visible; save language policy.	WF-18

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Localization | 2/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-L06	Emergency terminology: Keep terms consistent across channels; review translation editor.	WF-18

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Public address | 1/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-P01	Speaker zones & coverage: Reach people without a smartphone; select broadcast zones.	WF-21
FR-P02	Compose rescue announcement: Prepare an announcement for the right area; preview announcement.	WF-21
FR-P03	Preview & approve announcement: Check message, zones and playback behavior; approve sample playback.	WF-21
FR-P04	Speaker playback monitoring: Track playback separately from command receipt; inspect speaker health.	WF-21
FR-P05	Speaker device & self-test: SPK-A05-08 · playback unknown; open coverage fallback.	WF-21

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Public address | 2/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-P06	Live rescue microphone: Announce live to approved rescue zones; review live session.	WF-21
FR-P07	Drone / vehicle speaker mission: Coordinate mobile public-address support; review mission.	WF-21
FR-P08	Audio library & TTS review: Use approved speech in every supported language; edit voice content.	WF-21
FR-P09	PA priority & arbitration: Avoid competing announcements; save arbitration policy.	WF-21

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Care profiles | 1/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-H01	My rescue assistance profile: Tell responders what support you may need; review emergency sharing.	WF-26
FR-H02	Health & assistance sharing: Share only what the response team needs; save sharing preferences.	WF-26
FR-H03	Caregiver relationships: Review who can help manage your profile; review caregiver authority.	WF-26
FR-H04	Caregiver authorization: Verify authority before granting access; approve scoped relationship.	WF-26
FR-H05	Dependent person profile: Review the assistance plan with the person; request profile update.	WF-26

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Care profiles | 2/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-H06	Profile update & review: Keep assistance information current; submit profile update.	WF-26
FR-H07	People & assistance registry: Find people who may need proactive help; open assistance case.	WF-26
FR-H08	Proactive rescue queue: No SOS does not mean no need for help; review priority reasons.	WF-27
FR-H09	Priority evidence & decision: Review why assistance was recommended; assign proactive assistance.	WF-27
FR-H10	Presence & accountability: Distinguish presence from registration; open proactive queue.	WF-23

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Care profiles | 3/3

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-H11	Restricted medical record: Emergency medical subset · audited access; return to assistance.	WF-19
FR-H12	Welfare check outcome: Record contact even without an SOS; save welfare observation.	WF-27

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Intelligence | 1/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-Q01	Positioning source health: Know where location can be trusted; inspect location quality.	WF-23
FR-Q02	Position quality & history: Alex · position is evidence, not certainty; open assistance context.	WF-23
FR-Q03	Indoor calibration editor: Align anchors to a versioned floor plan; review route graph.	WF-23
FR-Q04	Evacuation optimization: Compare feasible evacuation strategies; review route comparison.	WF-24
FR-Q05	Route comparison & decision: Understand the trade-offs before publishing; publish guidance version.	WF-24

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

Intelligence | 2/2

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-Q06	AI incident copilot: A grounded summary with visible uncertainty; review AI recommendation.	WF-25
FR-Q07	AI evidence & approval: Approve an action, not a fluent paragraph; review route decision.	WF-25
FR-Q08	AI policies & model evaluation: Keep model behavior measurable; open model governance.	WF-25
FR-Q09	AI deployment governance: Publish a reviewed model and prompt version; save deployment draft.	WF-25
FR-Q10	Simulation & crowd scenarios: Test strategy without asserting real-world certainty; review route validation.	WF-24

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

PhysicalSystems | 1/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-F01	Physical systems overview: Never aggregate stale or impaired systems into a green all-clear; each zone retains its controller and observation time	WF-29
FR-F02	Controller inventory: Inventory discovery grants no write rights; bind exact model, firmware, site and interface evidence before enrollment	WF-28
FR-F03	Controller capability detail: Unknown or undocumented capability is read-only; certification and API compatibility are separate evidence	WF-28
FR-F04	Signal point mapping: Verify point polarity and meaning against wiring and witnessed tests; a valve contact is not proof of water flow	WF-28
FR-F05	Physical zone mapping: Reject ambiguous floor or cross-tenant zone mappings; require signed plan revision and site walkdown	WF-28

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

PhysicalSystems | 2/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-F06	Cause-and-effect matrix: Only the locally approved cause-and-effect matrix governs physical effects; cloud edits cannot silently rewrite controller logic	WF-28
FR-F07	Physical event detail: Deduplicate source sequence; retain raw evidence and distinguish alarm, supervisory, trouble, test and inferred incident	WF-29
FR-F08	Fire-water status: Pump-running, pressure, valve position and flow are independent signals; unknown water availability blocks readiness claims	WF-29
FR-F09	Controlled release review: Release is blocked without documented allowed operation, installed certification, fresh prerequisites, local permit and controller interlocks	WF-30
FR-F10	Command authorization: Bind distinct requester/approver to exact equipment, operation, parameters, zone, version and expiry; any change invalidates approval	WF-30

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

PhysicalSystems | 3/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-F11	Execution feedback: Controller acceptance is not physical completion; missing independent feedback becomes unknown with no blind retry	WF-30
FR-F12	Unknown-result reconciliation: Reconcile through supported readback and qualified observation; never resend a non-idempotent release to discover the outcome	WF-30
FR-F13	Local override & handover: Local override revokes remote authority; reconnection cannot replay commands or restore a revoked lease	WF-30
FR-F14	Impairment & test mode: Test mode requires bounded isolation and compensating measures; permit expiry escalates but does not assert restoration	WF-31
FR-F15	Commissioning plan: Freeze BOM, firmware, mappings and matrix; physical tests require site permit, witnesses, isolation and local abort	WF-32

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

PhysicalSystems | 4/4

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-F16	Test-run evidence: Expected results cannot populate observed results; failed or unexecuted tests block release and preserve attachments	WF-32
FR-F17	Compatibility & certificate dossier: A product certificate does not certify the installation or cloud adapter; exact scope, revisions and acceptance evidence must match	WF-32
FR-F18	Maintenance & restoration: Only witnessed restoration and site approval clear impairment; time elapsed or technician closure alone cannot restore availability	WF-31
FR-F19	Rescue equipment mission: Colossus control API is unverified: dispatch to trained vendor-console operator; no cloud movement, nozzle or casualty actuation	WF-33
FR-F20	Physical connector onboarding: Connector activation starts read-only; credentials stay in vault and only signed capability policy can enable an approved write operation	WF-28

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 1/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-001	Signal/Event: an immutable observation from a source, with observed_at and received_at.	System-wide
APP-SYS-002	Incident: an operational record collecting evidence, with severity and an owner.	System-wide
APP-SYS-003	SOS: a request for assistance for a person/group, with its own state and an optional incident link.	System-wide
APP-SYS-004	Warning: a message issued by version and audience.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 2/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-005	Delivery: one sending attempt through one channel.	System-wide
APP-SYS-006	Task: an assignment for a unit.	System-wide
APP-SYS-007	Personal status: a sourced statement/status, not an inference of absolute safety.	System-wide
APP-SYS-008	Command: an instruction to a device, separate from an event.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 3/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-009	Signal — received → validated → correlated / quarantined / rejected — Gateway validates schema/source/tenant; invalid signals do not enter live workflows	System-wide
APP-SYS-010	Incident — new → triaging → active → contained → resolved → closed — Operator triage; supervisor closes after checklist completion; reopening requires a reason	System-wide
APP-SYS-011	Incident exception — new/triaging → dismissed; any incident not yet closed → linked — Dismissal requires a reason; linking does not delete IDs or evidence	System-wide
APP-SYS-012	SOS — submitted → acknowledged → assigned → assisting → resolved — Server receipt differs from operator ACK; responders update only assigned tasks	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 4/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-013	SOS cancel — submitted/acknowledged/assigned → cancellation_requested → canceled or continued processing — After dispatch, an operator decision is required; citizens cannot delete the record	System-wide
APP-SYS-014	Person status — unknown, in_danger, help_requested, self_reported_safe, responder_confirmed_assisted — Append observations with timestamp/source; may return to danger; safe does not close the incident	System-wide
APP-SYS-015	Warning — draft → review → approved → sending → active → expired / canceled — New versions require reapproval; authors cannot approve their own work when policy requires separation	System-wide
APP-SYS-016	Task — offered → accepted → en_route → on_scene → completed — Offer deadline → expired → reassign; decline requires a reason; a task has one current owner	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 5/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-017	Command — requested → approved → dispatched → accepted → executing → succeeded / failed / unknown / expired — Device acceptance does not prove success; a timeout does not automatically mean physical failure	System-wide
APP-SYS-018	Physical asset: an installed, configuration-bound controller, pump, valve, releasing system, or rescue device.	System-wide
APP-SYS-019	Capability: independently approved observe/control permission for an exact operation.	System-wide
APP-SYS-020	Physical feedback: observed evidence distinct from transport/device acknowledgment.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 6/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-021	Impairment: an authorized reduction in protection requiring compensating measures and verified restoration.	System-wide
APP-SYS-022	Mission: a supervised equipment operation with local authority and handover.	System-wide
APP-SYS-023	UI label UNKNOWN maps to the existing machine state unknown; it is not a new terminal success/failure state.	System-wide
APP-SYS-024	Impairment lifecycle: requested → approved → active → restoration_pending → verified_closed; expiry without restoration stays restoration_pending and escalates.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 7/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-025	Mission lifecycle: draft → review → approved → locally_accepted → executing → completed / aborted / unknown; completion requires outcome evidence.	System-wide
APP-SYS-026	Every endpoint checks the principal, tenant membership, region, site scope, object ownership, and action permission on the server.	System-wide
APP-SYS-027	Hiding frontend buttons is not a security control.	System-wide
APP-SYS-028	Family members see only shared fields; operators see information needed for the incident; platform admins have no automatic data access.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 8/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-029	Warning approvers and command approvers have separate permissions.	System-wide
APP-SYS-030	Break-glass access requires a reason, expiry, audit, and supervisor notification; it does not bypass tenant boundaries.	System-wide
APP-SYS-031	Access to out-of-scope objects returns 404 or an error that does not disclose their existence.	System-wide
APP-SYS-032	Lists use cursor pagination, stable sorting (created_at,id), and a maximum of 100 records/page.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 9/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-033	All mutations require validation, idempotency when creating effects, and optimistic concurrency on updates.	System-wide
APP-SYS-034	List screens provide search/filter for appropriate fields and loading/empty/error/denied states; forms provide required/invalid/conflict/unsaved states; details provide stale/not-found states.	System-wide
APP-SYS-035	Prototype data is not real telemetry.	System-wide
APP-SYS-036	Physical control additionally checks the site/controller capability, vendor permission, OT access policy, current impairment/lockout, local authority, and operation-specific approval.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 10/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-037	The UI must show monitor-only, control eligible, inhibited, stale, and UNKNOWN distinctly.	System-wide
APP-SYS-038	Engineering access does not imply incident command authority.	System-wide
APP-SYS-039	Segment OT from public networks; allowlist supported endpoints and operations, use controlled maintenance access, and retain credential/change audit.	System-wide
APP-SYS-040	Local automatic detection, pump starting, and releasing protection must not depend on cloud approval or connectivity.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 11/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-041	An admin creates a tenant with an immutable home_region; selects locale/timezone, configures SSO/MFA, and invites memberships by site.	WF-01
APP-SYS-042	Users select a workspace before entering the console.	WF-01
APP-SYS-043	Recovery returns the same message for existing and nonexistent emails to reduce enumeration.	WF-01
APP-SYS-044	Invitations are single-use, have a TTL, and enforce email/domain policy.	WF-01

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 12/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-045	When SSO is unavailable, a controlled break-glass account is available within the appropriate tenant; MFA is not automatically bypassed.	WF-01
APP-SYS-046	AC: expired invitations are rejected; revoked roles cannot use old tokens after the proposed 60 s revocation propagation limit; switching workspaces does not display the previous tenant's cache.	WF-01
APP-SYS-047	Site admins create a country/region/site/building/floor/zone/road-segment/geofence graph.	WF-02
APP-SYS-048	Not every site must have floors.	WF-02

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 13/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-049	Device binding checks that the device belongs to the same tenant.	WF-02
APP-SYS-050	Floor plans have revisions, scale, exits, accessibility, and an approver.	WF-02
APP-SYS-051	Each incident stores the plan revision used.	WF-02
APP-SYS-052	Locations carry method, accuracy, and age; GPS does not automatically infer a floor.	WF-02

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 14/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-053	Without positioning, users can enter a location manually and send an SOS.	WF-02
APP-SYS-054	AC: plans without a reviewer cannot be published; zones with active incidents cannot be hard-deleted; routes without freshness information show “not verified” and are not presented as certainly safe.	WF-02
APP-SYS-055	Adapters check credentials, tenant binding, schema, clock skew, and payload limits, then durably store the event before issuing a receipt.	WF-03
APP-SYS-056	A duplicate key (tenant,source,event_id) returns the previous receipt.	WF-03

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 15/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-057	Late events are retained for audit, but live rules evaluate TTL by hazard.	WF-03
APP-SYS-058	Unknown sources/permissions/units/locations are quarantined.	WF-03
APP-SYS-059	Inputs support HTTPS/CloudEvents, MQTT adapters, CAP, camera metadata, wearables, telematics, and manual entry.	WF-03
APP-SYS-060	AC: replaying the same ID 100 times creates one event; two tenants using the same event ID do not collide; sources spoofing a tenant are blocked; stale events do not initiate commands.	WF-03

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 16/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-061	Versioned rules provide candidate incidents by time window, zone, hazard, and source.	WF-04
APP-SYS-062	Operators review evidence and confidence, then create or link incidents; splitting preserves audit records and source links.	WF-04
APP-SYS-063	Different hazards are not combined merely because they are nearby; cross-hazard relationships have a type (caused_by/nearby/possible_duplicate).	WF-04
APP-SYS-064	Camera detection is labeled “possible fire”; fire panel policy may immediately initiate a response through a site-approved workflow without waiting for AI.	WF-04

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 17/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-065	AC: merging does not lose SOS/tasks; splitting does not duplicate dispatch; rule changes do not rewrite history; dismissal requires a reason.	WF-04
APP-SYS-066	Users select a category or Other/Unknown; enter/update location, notes, and the number of people.	WF-05
APP-SYS-067	The emergency flow does not require a complete profile.	WF-05
APP-SYS-068	The server returns an SOS ID and received_at after commit.	WF-05

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 18/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-069	The client displays pending when the request is only stored locally.	WF-05
APP-SYS-070	Resending uses client_request_id, TTL, and reconciled state.	WF-05
APP-SYS-071	Threat/discreet requests default to no sound and no group fan-out; ordinary assistance uses rule-based severity rather than treating every SOS as P1.	WF-05
APP-SYS-072	AC: network loss does not show “received”; resending does not create a second SOS; users who deny GPS can still enter location manually; changing category does not lose the ID.	WF-05

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 19/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-073	An approved detector emits possible_fall; the app displays a proposed 30 s check-in with “I'm OK” and “Get help now”.	WF-06
APP-SYS-074	Without a response, the server timer continues even if the app closes, creating an operator/caregiver request according to consent.	WF-06
APP-SYS-075	An offline device produces unknown status, not a conclusion that the person fell.	WF-06
APP-SYS-076	Check-in duration is configurable by service/group, with an accessible alternative; it is not automatically used as a diagnosis.	WF-06

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 20/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-077	AC: escalation continues after the app is killed; a late “I'm OK” updates the record without silently withdrawing an ongoing task; false positives retain a review reason.	WF-06
APP-SYS-078	Reports/telematics provide position, heading if available, road segment, vehicle count, and injuries=unknown.	WF-07
APP-SYS-079	Correlation distinguishes opposite traffic directions and bridge levels.	WF-07
APP-SYS-080	Users can submit a minimal quick report and add details later.	WF-07

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 21/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-081	Dispatch targets configured units; without an agreement, the system does not automatically call a public emergency endpoint.	WF-07
APP-SYS-082	AC: missing injury data is accepted as unknown; one acceleration sample does not auto-confirm a collision; sensitive data goes only to the assigned team.	WF-07
APP-SYS-083	Connectors allowlist issuers and validate XML/schema, identifiers, sent, status, scope, info/area, and expiry.	WF-08
APP-SYS-084	Deduplicate sender+identifier; Update/Cancel references the previous version and must preserve lineage.	WF-08

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 22/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-085	Official warnings and local responses are separate objects.	WF-08
APP-SYS-086	Expired warnings do not continue live broadcast; cancellation does not delete previously sent versions.	WF-08
APP-SYS-087	Polygons use CAP's coordinate order and are explicitly converted to GeoJSON; lat/lon must not be implicitly swapped.	WF-08
APP-SYS-088	AC: Test/Exercise does not enter the Actual flow; unauthorized issuers are quarantined; Update replaces the active version while retaining the previous one; users outside the polygon are not selected because of coordinate errors.	WF-08

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 23/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-089	Authors select the incident, area/site/group, severity, languages, content, and expiry.	WF-09
APP-SYS-090	Audience preview is a snapshot of eligible recipients and exclusion reasons; an independent approver signs the version.	WF-09
APP-SYS-091	Changing content/audience after approval requires reapproval.	WF-09
APP-SYS-092	Rapid break-glass issuance requires special permission and audit; not every P1 must wait for approval if the site playbook permits preapproved templates.	WF-09

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 24/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-093	Machine translation is draft-only; safety content must be reviewed.	WF-09
APP-SYS-094	AC: approval of v3 does not permit sending v4; an audience ID targeting the wrong tenant is blocked; scheduled warnings recheck expiry and permissions at issuance.	WF-09
APP-SYS-095	Notifications create individual delivery attempts by recipient/channel/version; queued/submitted/provider_accepted/delivered/read/human_ack/failed/unknown remain distinct states.	WF-10
APP-SYS-096	Provider callbacks are authenticated and replay-protected.	WF-10

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 25/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-097	An ACK deadline triggers durable escalation with an owner and maximum steps.	WF-10
APP-SYS-098	Channel fallback does not automatically send private content by SMS; SMS templates are minimal and use authenticated links when needed.	WF-10
APP-SYS-099	Push notifications cannot guarantee bypassing the operating system's silent mode.	WF-10
APP-SYS-100	AC: provider acceptance does not increase the human ACK count; duplicate callbacks do not duplicate escalation; provider failures use bounded retries and alternate providers, and 429 respects retry_after.	WF-10

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 26/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-101	Dispatchers filter units by capability, availability, site authorization, and latest location.	WF-11
APP-SYS-102	ETA is only a timestamped estimate.	WF-11
APP-SYS-103	Offers have a deadline; units accept atomically using a version.	WF-11
APP-SYS-104	If nobody accepts, escalate; do not automatically mark dispatch as completed.	WF-11

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 27/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-105	Responders transition through en_route/on_scene, record observations and handover, and complete tasks; task completion does not close the incident.	WF-11
APP-SYS-106	AC: two dispatchers cannot assign two owners to one task; offline acceptance must remain pending until server confirmation; reassignment retains the previous responder's notes as observations.	WF-11
APP-SYS-107	Safety reports store status, person, incident scope, observed_at, reporter, and location precision.	WF-12
APP-SYS-108	Fan-out reaches all eligible subscribed groups at report time, plus verified contacts; recipients are deduplicated.	WF-12

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 28/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-109	Each group receives status; detailed location requires specific consent.	WF-12
APP-SYS-110	Revoked memberships/consent are rechecked before delivery.	WF-12
APP-SYS-111	Group chat includes membership, abuse reporting, and individual blocking for social interaction; blocking cannot disable mandatory site warnings under the published policy.	WF-12
APP-SYS-112	AC: one contact in 3 groups does not receive 3 identical SMS messages; location consent off does not leak coordinates; a safe report does not close a building incident; unknown is not counted as safe.	WF-12

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 29/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-113	Operators maintain shelter status, estimated capacity, timestamp, accessibility, and valid_until.	WF-13
APP-SYS-114	Offline packs store versioned, expiring plans/guidance/contacts and do not guarantee live routes.	WF-13
APP-SYS-115	Voice uses the same approved text with a transcript and pause/replay; autoplay must not expose discreet mode.	WF-13
APP-SYS-116	Without a verified route, display only the plan and approved guidance rather than automatically drawing an optimized fire-avoidance route.	WF-13

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 30/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-117	AC: a shelter past valid_until becomes unknown; missing capacity data is not shown as 0; offline plans display their age.	WF-13
APP-SYS-118	Enrollment tokens are single-use; device certificates support rotation/revocation.	WF-14
APP-SYS-119	Heartbeat thresholds depend on device class; stale/disconnected is distinct from the area's safety status.	WF-14
APP-SYS-120	Maintenance suppression has a start/end, reason, approver, and replacement coverage; blind spots are always visible.	WF-14

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 31/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-121	When maintenance ends, monitoring automatically resumes and is logged.	WF-14
APP-SYS-122	AC: revoked certificates cannot ingest; suppression does not continue after maintenance ends; firmware upgrades do not automatically run fleet-wide during an active incident.	WF-14
APP-SYS-123	Physical maintenance additionally follows WF-31: ending a software suppression window does not prove that physical protection has been restored; required local checks and acknowledgment remain mandatory.	WF-14
APP-SYS-124	Commands are limited to declared capabilities; parameter schema, zone, expiry, interlock, authorized approver, and local controller policy are all mandatory.	WF-15

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 32/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-125	The web interface cannot send arbitrary raw commands.	WF-15
APP-SYS-126	Gateway receipt, device accepted, executing, and verified result are separate.	WF-15
APP-SYS-127	An unknown timeout does not blindly retry a non-idempotent command.	WF-15
APP-SYS-128	Cloud actuation does not replace certified local protection loops.	WF-15

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 33/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-129	Closed-head sprinklers typically activate thermally; the prototype does not assume software can open individual sprinkler heads.	WF-15
APP-SYS-130	AC: commands past TTL do not execute after reconnect; zone mismatches are rejected; acceptance without feedback remains unknown/awaiting confirmation; simulators have no live credentials.	WF-15
APP-SYS-131	WF-30 refines this command workflow for installed suppression and rescue equipment.	WF-15
APP-SYS-132	Wet-pipe monitoring, releasing-controller requests, pump requests, and mobile equipment actions remain separate capabilities; use WF-28 binding and WF-32 acceptance before enabling any control.	WF-15

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 34/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-133	Fleet adapters primarily provide telemetry/media and health.	WF-16
APP-SYS-134	A mission draft includes purpose, incident, route/geofence, altitude reference, pilot, clearance, weather check, and lost-link policy.	WF-16
APP-SYS-135	The local pilot decides deployment according to the integrated system's capabilities.	WF-16
APP-SYS-136	Approval does not mean takeoff has occurred.	WF-16

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 35/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-137	AI detectors do not automatically issue TAKEOFF.	WF-16
APP-SYS-138	Abort/return commands require capability support and flight-stack confirmation; the UI does not promise universal cancellation.	WF-16
APP-SYS-139	AC: no approval without pilot/clearance; stale telemetry disables position-dependent commands; link loss shows unknown while local policy continues; mission evidence has hash/provenance.	WF-16
APP-SYS-140	Install manifest → configure secret reference → map → sandbox test → approve → activate → monitor → rotate/deprecate.	WF-17

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 36/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-141	Connectors run with least privilege, outbound allowlists, quota/backpressure, and a kill switch.	WF-17
APP-SYS-142	DLQ replay requires a reason, a valid schema, and retention of the original ID; command replay is blocked by default.	WF-17
APP-SYS-143	Cross-tenant exchange requires an agreement and payload allowlist.	WF-17
APP-SYS-144	AC: adapter failures do not block direct SOS; oversized payloads return 413; incompatible major schemas are quarantined; secrets do not appear in logs/exports.	WF-17

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 37/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-145	Use UTC for storage, IANA timezones for display/schedules, BCP47 locales, Unicode, and E.164 phone numbers according to country.	WF-18
APP-SYS-146	The UI includes RTL layout tests, pluralization, and pseudo-localization.	WF-18
APP-SYS-147	Do not hardcode one worldwide emergency number; the emergency contact registry is verified by country/site.	WF-18
APP-SYS-148	Residency covers databases, object storage, backups, logs, support, and exports; global routing does not carry raw SOS data.	WF-18

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 38/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-149	AC: schedules crossing DST do not send twice, using a schedule occurrence ID; long locales do not obscure CTAs; changing region is a reviewed migration, not merely editing a dropdown.	WF-18
APP-SYS-150	Data classes have retention policies, legal holds, and deletion jobs with reports.	WF-19
APP-SYS-151	Export/deletion verifies the requester, scope, and exceptions; held evidence is not deleted.	WF-19
APP-SYS-152	Audit is append-only with actor, action, object/version, reason, time, correlation, and before/after digests.	WF-19

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 39/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-153	Do not embed tokens, raw secrets, or complete medical notes in audit records.	WF-19
APP-SYS-154	AC: precise-location access is logged; deletion does not remove held records; exports exclude other tenants; break-glass access automatically revokes at TTL expiry.	WF-19
APP-SYS-155	Closure requires an open SOS/task check, disposition of people with unknown status, partner handover, a reason, and a supervisor.	WF-20
APP-SYS-156	An “All clear” warning is a separate action.	WF-20

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 40/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-157	Drills require environment+mode isolation in the broker, credentials, and recipient directory; a banner alone is insufficient.	WF-20
APP-SYS-158	Reports separate transport latency, operator latency, and response latency; samples with missing data explicitly state their denominator.	WF-20
APP-SYS-159	AC: safety check-ins do not close incidents; drills do not call live webhooks/actuators; shift handovers have incoming acknowledgment; reports are deidentified when the user lacks PII permission.	WF-20
APP-SYS-160	Bind the installed regional model, serial number, firmware, gateway, license, vendor permission, and compatibility dossier before activating an adapter.	WF-28

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 41/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-161	Import and independently review point addresses, semantic states, units/scaling, site/zone associations, and cause-and-effect revisions.	WF-28
APP-SYS-162	Distinguish wet-pipe, pre-action/deluge, pump, and mobile rescue assets.	WF-28
APP-SYS-163	Unknown capability defaults to monitor-only; onboarding cannot create control authority.	WF-28
FR-PHY-01	FR-PHY-01: retain the exact equipment/interface identity and approved evidence revision.	WF-28

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 42/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-PHY-02	FR-PHY-02: validate point and zone mapping against physical labels and reject cross-tenant, duplicate, or unresolved mappings.	WF-28
FR-PHY-03	FR-PHY-03: version the cause-and-effect and read/control capability contracts; changes require impact review and reacceptance where affected.	WF-28
APP-SYS-167	AC: an unbound firmware or license change suspends dependent control; a wrong-zone mapping cannot be activated; wet-pipe heads are not offered as individually controllable outputs.	WF-28
APP-SYS-168	Ingest manufacturer-supported panel, pump, valve, release, waterflow, power, and link states with provenance and quality.	WF-29

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 43/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-169	Preserve controller alarm priority and local protection when cloud/edge is unavailable.	WF-29
APP-SYS-170	Separate demand, motor running, release output, valve position, waterflow, and confirmed field observations.	WF-29
APP-SYS-171	Missing or stale values remain unknown; do not turn a link fault into a safe indication.	WF-29
FR-PHY-04	FR-PHY-04: store source/receive time, source address, unit, quality, mapping revision, and raw evidence reference for each observation.	WF-29

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 44/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-PHY-05	FR-PHY-05: supervise freshness/link health using approved device/site thresholds and expose coverage gaps and compensating measures.	WF-29
FR-PHY-06	FR-PHY-06: correlate without suppressing local protective actions; deduplicate transport replay while retaining state transitions and restoration evidence.	WF-29
APP-SYS-175	AC: stale pump-running data is visibly uncertain; pressure alone does not confirm discharge; loss of cloud leaves local automatic protection available; duplicate telemetry does not create duplicate incidents.	WF-29
APP-SYS-176	Only a site-accepted, explicitly control-capable operation may create a physical request.	WF-30

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 45/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-177	Validate actor authority, incident/site/zone, approved parameter envelope, current physical readiness, local interlocks, command expiry, and the policy's independent approval requirements.	WF-30
APP-SYS-178	Site/controller authority prevails.	WF-30
APP-SYS-179	The system never assumes that BACnet/Modbus writes, a relay, or a network ACK authorizes or proves suppression.	WF-30
APP-SYS-180	AI cannot originate autonomous release/start/motion actions.	WF-30

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 46/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-PHY-07	FR-PHY-07: reject monitor-only or undocumented operations server-side and at the edge; preserve the approved request and command-policy version.	WF-30
FR-PHY-08	FR-PHY-08: enforce TTL, unique request ID, duplicate-effect prevention, and bounded delivery; do not replay expired requests after reconnect or blindly retry non-idempotent actions.	WF-30
FR-PHY-09	FR-PHY-09: distinguish transport receipt, device acceptance, execution, and physical feedback; evaluate success only against operation-specific approved evidence.	WF-30
FR-PHY-10	FR-PHY-10: display UNKNOWN for missing/conflicting outcomes; assign reconciliation to a named operator and support local override and acknowledged manual handover without erasing history.	WF-30

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 47/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-185	AC: ACK without required feedback cannot become succeeded; duplicate/expired requests produce no new effect; loss of feedback opens reconciliation; local override blocks conflicting remote control.	WF-30
APP-SYS-186	An authorized impairment identifies equipment, affected zones, time window, reason, owner, compensating protection, and the local test/lockout arrangement.	WF-31
APP-SYS-187	Keep test and production credentials, routes, and outputs distinct; a UI banner is insufficient.	WF-31
APP-SYS-188	Expiry prompts restoration verification and escalation rather than claiming protection has returned.	WF-31

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 48/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-189	Local safety ownership remains explicit during shift handover.	WF-31
FR-PHY-11	FR-PHY-11: require approved impairment and physical isolation/lockout evidence before relevant tests; block unauthorized production commands while preserving agreed local protection arrangements.	WF-31
FR-PHY-12	FR-PHY-12: close an impairment only after required physical state checks, restoration evidence, and responsible-person acknowledgment; track overdue restoration and residual gaps.	WF-31
APP-SYS-192	AC: a simulated test cannot reach a live output; expiration without verification remains impaired/awaiting restoration; maintenance handover has an incoming owner.	WF-31

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 49/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-193	Maintain a configuration-specific dossier of manufacturer documents, interface permissions, regional listings/limitations, point maps, electrical/hydraulic dependencies, approved cause-and-effect, FAT, SAT, commissioning, calibration, and operator training.	WF-32
APP-SYS-194	Manufacturer research and simulator results are not field acceptance.	WF-32
APP-SYS-195	No whole-system certification is inferred from a component listing.	WF-32
FR-PHY-13	FR-PHY-13: track document revision, provenance, reviewer, expiry where applicable, and acceptance scope per model/firmware/site/capability.	WF-32

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 50/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
FR-PHY-14	FR-PHY-14: gate production capability activation on signed required tests and approvals; equipment, firmware, mapping, or policy changes trigger impact assessment and invalidate affected acceptance until revalidated.	WF-32
APP-SYS-198	AC: missing SAT blocks production control even after simulator PASS; an unrelated site's certificate cannot satisfy the gate; changed firmware leaves unaffected read capabilities available only if their contract remains valid.	WF-32
APP-SYS-199	A mission binds the actual supported robot or water-monitor interface, purpose, incident, qualified operator, operating area, route/no-go zones, water/energy supply, telemetry freshness, and lost-link/local-emergency policy.	WF-33
APP-SYS-200	Observe-only equipment remains observe-only.	WF-33

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 51/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-201	Approval is not movement or discharge.	WF-33
APP-SYS-202	The local operator determines deployment and remains able to take over according to the vendor-supported design; no universal stop/return function is promised.	WF-33
FR-PHY-15	FR-PHY-15: require mission-specific authority, capability evidence, operating envelope, and readiness checks; separate movement, aiming, and water discharge permissions.	WF-33
FR-PHY-16	FR-PHY-16: record telemetry and action feedback, enforce approved lost-link behavior locally, and provide acknowledged manual handover and outcome evidence; never infer rescue completion from command acceptance.	WF-33

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 52/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-205	AC: missing vendor-supported control or operator blocks dispatch of physical actions; stale position blocks dependent aiming/motion; link loss shows UNKNOWN and the approved local policy; a handover retains responsibility and evidence.	WF-33
NFR-01	NFR-01 — Production regional core 99.95% monthly — Synthetic + server metrics; include dependency failures affecting users	System-wide
NFR-02	NFR-02 — Ingest p95 ≤ 2 s; fan-out enqueue p95 ≤ 5 s — Load of 100 event/s steady, 1,000 event/s for 60 s; 10,000 recipients/incident; metadata payload ≤ 64 KB	System-wide
NFR-03	NFR-03 — Priority fairness — P1 reserved queues; tenant noisy-neighbor test; bulk telemetry must not block SOS	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 53/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
NFR-04	NFR-04 — Multi-AZ target RPO 0 for committed records, RTO ≤ 15 minutes — Kill primary, verify durable commit and failover; commit only after testing	System-wide
NFR-05	NFR-05 — Regional disaster RPO ≤ 5 minutes, RTO ≤ 60 minutes — Restore/cold-region drill within residency policy; immutable backup restore	System-wide
NFR-06	NFR-06 — WCAG 2.2 AA target — Keyboard, screen reader, zoom, contrast, and error announcements; no certification claim yet	System-wide
NFR-07	NFR-07 — Web interactive p95 ≤ 3 s on the reference network — 4G simulation, representative low-end device; map loads separate from core content	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 54/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
NFR-08	NFR-08 — Offline durable outbox and stale visibility — Airplane mode, app kill, reconnect, clock skew, and reassignment conflict	System-wide
NFR-09	NFR-09 — Security isolation — Tenant/object permission tests, upload validation, credential revocation, audit coverage	System-wide
NFR-10	NFR-10 — Backward compatibility — API/event contract tests; additive minor changes; parallel major versions with sunset	System-wide
APP-SYS-216	Reference loads are engineering targets, not measured benchmarks.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 55/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-217	Media uploads use object storage separately from the critical event pipeline.	System-wide
APP-SYS-218	NFRs apply to the product to be built; the static prototype does not demonstrate that these requirements are met.	System-wide
NFR-11	NFR-11 — Local automatic protection independent of cloud/edge availability — Witness cloud/link/adapter failure without inhibiting the approved local protection sequence	System-wide
NFR-12	NFR-12 — Configuration-specific telemetry/feedback timing and quality — Signed freshness, delivery, and feedback deadlines per device/site; measure separately; UNKNOWN on deadline violation	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 56/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
NFR-13	NFR-13 — OT and command isolation — Prove monitor-only enforcement, denied writes, least privilege, maintenance access control, and no test-to-live path	System-wide
NFR-14	NFR-14 — Configuration-specific physical assurance — Complete provenance and signed required FAT/SAT; changes invalidate affected capability acceptance	System-wide
APP-SYS-223	These physical NFRs do not substitute software latency targets for physical response or hydraulic performance.	System-wide
APP-SYS-224	Device/site timing and pass criteria must be approved before tests, and results must identify firmware, hardware, and operating conditions.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 57/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-225	Run complete Fire, Disaster, Fall, Road, discreet SOS, and generic assistance flows; test duplicates, late data, out-of-order data, expiry, network partitions, unauthorized tenants, quota saturation, provider failures, and restore.	System-wide
APP-SYS-226	For physical/drone integrations, simulator PASS does not replace on-site acceptance.	System-wide
APP-SYS-227	The detailed matrix is in Acceptance-Matrix.csv.	System-wide
APP-SYS-228	Screen-level acceptance is in Screen-Specifications.md and linked to each mockup ID.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 58/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SYS-229	The physical gate additionally executes PHY-01–PHY-16 and AC-F01–AC-F20, checks every FR-PHY-01–FR-PHY-16 requirement, and requires the approved physical test arrangement and authorized site team.	System-wide
APP-SYS-230	Acceptance-Matrix.csv contains 65 cross-system and 241 screen cases (298 total).	System-wide
APP-SYS-231	Designed cases are not execution evidence.	System-wide
APP-SYS-232	Supplier support, physical permits, local operator readiness, and signed FAT/SAT are dependencies for enabling production control; missing evidence leaves the relevant capability unavailable or monitor-only.	System-wide

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 59/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SUP-01	Orchestrate PA using exact message/zone digest, priority, repeat and expiry; arbitrate competing zone leases.	WF-21
APP-SUP-02	Separate controller ACK, buffering/playing/completed and acoustic evidence; faults create coverage gaps/fallback.	WF-21
APP-SUP-03	Live mic needs explicit action, approved zones, short-lived tokens, indicators, heartbeat and timeout stop.	WF-21
APP-SUP-04	Media gateway bridges WebRTC to supported vendor protocols; isolate OT and cache checksummed/versioned audio.	WF-21

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 60/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SUP-05	Preserve local fire priority; scope stop to one broadcast; never replay expired messages after reconnect.	WF-21
APP-SUP-06	Validate floor/heading quality; uncertain position uses landmarks/SOS, never invented left/right guidance.	WF-22
APP-SUP-07	Exclude forbidden/blocked/unknown-critical edges and expired destinations; accessibility is a hard constraint.	WF-22
APP-SUP-08	Dijkstra/A* selects feasible routes only; penalties never bypass exclusions; no path means assistance/refuge.	WF-22

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 61/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SUP-09	Withdraw hazard-cut routes immediately; replan with hysteresis, cancel old TTS; stale cache uses approved procedures.	WF-22
APP-SUP-10	Align vi/en/ja text/voice, lexicon, pronunciation, transcript and visual/haptic alternatives.	WF-22
APP-SUP-11	Route API returns LOCATION_UNCERTAIN/NO_VERIFIED_ROUTE when unverified; arrival never automatically marks safe.	WF-22
APP-SUP-12	Ingest GNSS/QR/NFC/BLE/UWB/Wi-Fi/manual by capability/permission, calibration and coordinate frame.	WF-23

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 62/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SUP-13	Fuse floor hypotheses/outliers/motion/quality; never average floors or discard provenance on correction.	WF-23
APP-SUP-14	Separate registered/expected/observed/last-seen/absent/unknown presence; beacon loss never means absent/safe.	WF-23
APP-SUP-15	Produce one primary and up to two alternatives with reasons/assumptions/validity; diversify bottlenecks where feasible.	WF-24
APP-SUP-16	Balance cohorts/capacity using uncertain occupancy/throughput; regional strategy changes need operator approval.	WF-24

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 63/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SUP-17	Keep simulations separate from observed hazards; show assumptions and ETA intervals/confidence, not false precision.	WF-24
APP-SUP-18	AI retrieves approved SOPs and returns structured evidence/cutoff/uncertainty; unsupported claims cannot instruct.	WF-25
APP-SUP-19	AI only compares planner routes read-only; no graph edits, warning approval, commands, diagnosis or incident closure.	WF-25
APP-SUP-20	Policy proxy blocks injection/excess scope; regional models use minimal health data and redacted logs, no chain-of-thought.	WF-25

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 64/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SUP-21	Evaluate grounding/injection/leakage/unsafe action/locale/abstention before promotion; review feedback before training.	WF-25
APP-SUP-22	AI timeouts cannot block alarms/SOS/dispatch; streaming text cannot reach PA before validation/approval.	WF-25
APP-SUP-23	Separate identity from functional-needs profiles; preserve source/review/consent; self-report is not medically verified.	WF-26
APP-SUP-24	Verify guardians/caregivers with revocable field scopes; dispatchers/responders receive only necessary emergency subsets.	WF-26

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 65/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SUP-25	Audit health access and scoped/time-bound/reasoned break-glass; social circles cannot see medical notes/impairments.	WF-26
APP-SUP-26	Build candidates from presence/impact/needs/no confirmation; uncertain presence triggers welfare checks, not asserted location.	WF-27
APP-SUP-27	Explain priority by exposure/ability/isolation/route/contact/confidence; exclude wealth/nationality and age alone.	WF-27
APP-SUP-28	Deduplicate ProactiveCase by incident/person/policyVersion; later SOS links the case without duplicate dispatch.	WF-27

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

Background capabilities and system requirements | 66/66

Requirement IDs trace this proposal to detailed design and acceptance criteria.

ID	FUNCTION / OUTCOME	TRACE
APP-SUP-29	Escalate non-response without confirming injury; location corrections retain evidence/audit and trigger reevaluation.	WF-27

Defined design scope · Conditions, exceptions and acceptance: FRD + Screen Specifications.

SmartAlarm

BMS · Business workflows · 1/1

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

WF-34	BMS onboarding and mapping approval
WF-35	BMS alarm intake and reconciliation
WF-36	Governed acknowledgement and command requests
WF-37	Connector operation, upgrade and handover

Implementation design scope; vendor capabilities and physical effects require site acceptance.

SmartAlarm

BMS · Every additional screen · 1/2

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS01	BMS identity and capabilities
FR-BMS02	Semantic topology
FR-BMS03	Cross-system alarm reconciliation
FR-BMS04	Control authority and priority

Implementation design scope; vendor capabilities and physical effects require site acceptance.

SmartAlarm

BMS · Every additional screen · 2/2

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS05	Point drift and impact
FR-BMS06	Budgets and data freshness
FR-BMS07	Edge recovery and handover
FR-BMS08	Exchange tracing and loop control

Implementation design scope; vendor capabilities and physical effects require site acceptance.

SmartAlarm

BMS · Every detailed function · 1/10

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS-01	Site BMS inventory. Activation requires build, interface and evidence.
FR-BMS-02	Capability-specific access. Read, history, ACK and commands have separate server permissions.
FR-BMS-03	Bounded discovery. Scan approved scope only; cancellable; alarm intake remains available.
FR-BMS-04	Point and object import. Preview row errors, duplicate IDs and encoding; commit a revision.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

BMS · Every detailed function · 2/10

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS-05	Unit and semantic mapping. Golden fixtures verify scale, enums, types and byte order.
FR-BMS-06	Equipment and zone topology. Distinguish room, HVAC zone, fire compartment and route edge.
FR-BMS-07	Mapping publication and impact review. Show dependencies and approve revisions; rollback does not undo physical effects.
FR-BMS-08	Observation quality and freshness. Zero differs from missing; late data cannot overwrite newer observations.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

BMS · Every detailed function · 3/10

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS-09	Subscriptions and polling. Renew/reconcile lost subscriptions; polling follows measured budgets.
FR-BMS-10	History and investigation. Preserve UTC, timezone, gaps and raw/interpolated distinctions.
FR-BMS-11	Native alarm lifecycle. Active, ACK, confirmed and suppressed are independent; unsupported=null.
FR-BMS-12	Priority and severity mapping. Do not invert priority numbers globally; test per source.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

SmartAlarm

BMS · Every detailed function · 4/10

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS-13	Correlation and provenance. A BMS mirror and direct panel feed are not independent evidence.
FR-BMS-14	Controlled external ACK. Target the correct occurrence/version; failed receipt remains pending.
FR-BMS-15	Observe external acknowledgement. Preserve source actor; ACK does not close the rescue incident.
FR-BMS-16	Publish incident summaries to BMS. Approved namespace, minimal data and anti-echo; no medical records.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

BMS · Every detailed function · 5/10

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS-17	Operator alarm grouping. Retain raw events and grouping counts; unacknowledged P1 remains visible.
FR-BMS-18	Maintenance and suppression. Require reason, owner and expiry; elapsed maintenance does not prove restoration.
FR-BMS-19	BMS evidence for evacuation. Stale door/fan/lift evidence lowers confidence or withdraws guidance.
FR-BMS-20	Proactive rescue. Combine needs and fresh presence; occupancy counts are not identities.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

SmartAlarm

BMS · Every detailed function · 6/10

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS-21	Approved PA and app response. Do not wait for an LLM; echo/history cannot repeat broadcasts.
FR-BMS-22	Control impact preview. Show target, owner, priority, preconditions and affected zones.
FR-BMS-23	Approved commands with expiry. Parameter changes invalidate approval; local override retains priority.
FR-BMS-24	Priority lease and relinquishment. Use permitted slots only; never clear an unproven foreign slot.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

SmartAlarm

BMS · Every detailed function · 7/10

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS-25	Physical outcome reconciliation. Separate transport receipt, controller report and independent feedback.
FR-BMS-26	Timeout and unknown outcomes. Do not resend unknown effects; query/read back first.
FR-BMS-27	Local override and handover. Do not restore old commands on reconnect; assign cleanup ownership.
FR-BMS-28	Durable journal and offline operation. Do not ACK before durable commit; local protection remains independent.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

BMS · Every detailed function · 8/10

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS-29	HA and single writer. Fence at the actual dispatch point; standby cannot write.
FR-BMS-30	Alarm resynchronization after gaps. Uncertain snapshot/cursor exposes gaps; one page cannot clear missing alarms.
FR-BMS-31	Schema drift and upgrades. Rename preserves identity; type/unit changes quarantine affected capabilities.
FR-BMS-32	License, certificate and account lifecycle. Test rotation; revocation blocks privileges; never log secrets.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

BMS · Every detailed function · 9/10

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS-33	Data- and zone-aware health. Online gateway does not hide stale points; alarm latency is measured separately.
FR-BMS-34	Capacity, backfill and retention. Per-BMS quotas; historian work cannot starve alarm queues.
FR-BMS-35	Actual and exercise isolation. Simulation never sends real ACK/writes; commissioning has a separate plan.
FR-BMS-36	Configuration-specific FAT/SAT. Sign off per capability; passing reads does not authorize commands.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

SmartAlarm

BMS · Every detailed function · 10/10

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

FR-BMS-37	Support and forensic export. Timeline includes revisions/sources; redact secrets and health data.
FR-BMS-38	Evidence-based BMS AI assistance. LLM reads scoped, sourced evidence; it never controls directly.
FR-BMS-39	Tenant/site isolation and residency. Repeated native IDs remain isolated; cross-site access uses authorized projections.
FR-BMS-40	Controlled retirement and handover. Review open alarms/leases, hand over, revoke, verify and retain audit.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

SmartAlarm

BMS · Every connector profile · 1/3

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

CON-BMS-01	bms.alc.webctrl-events — version/licence-specific profile; activate capabilities after qualification.
CON-BMS-02	bms.alc.webctrl-soap-sdk — version/licence-specific profile; activate capabilities after qualification.
CON-BMS-03	bms.delta.enteliweb — version/licence-specific profile; activate capabilities after qualification.
CON-BMS-04	bms.distech.ecnet-niagara — version/licence-specific profile; activate capabilities after qualification.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

SmartAlarm

BMS · Every connector profile · 2/3

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

CON-BMS-05	bms.honeywell.ebi-api — version/licence-specific profile; activate capabilities after qualification.
CON-BMS-06	bms.honeywell.webs-niagara — version/licence-specific profile; activate capabilities after qualification.
CON-BMS-07	bms.jci.metasys-rest — version/licence-specific profile; activate capabilities after qualification.
CON-BMS-08	bms.schneider.ebo-ews — version/licence-specific profile; activate capabilities after qualification.

Implementation design scope; vendor capabilities and physical effects require site acceptance.

SmartAlarm

BMS · Every connector profile · 3/3

Trace directly to FRD, Detail Design, Screen Specifications and connector profiles.

CON-BMS-09	bms.siemens.desigo-wsi — version/licence-specific profile; activate capabilities after qualification.
CON-BMS-10	bms.tridium.nds-cloud — version/licence-specific profile; activate capabilities after qualification.
CON-BMS-11	bms.tridium.niagara-local — version/licence-specific profile; activate capabilities after qualification.

Implementation design scope; vendor capabilities and physical effects require site acceptance.